Adaptive Heuristic Ecosystem for Online Multi-Robot Task Allocation under Failures, Mixed Stress, and Deadline Pressure

Abstract

Online multi-robot task allocation must keep producing sound assignments while tasks arrive, deadlines tighten, and robots fail in mid-execution. No single allocation rule serves all of these regimes: a policy tuned for spatial efficiency loses deadlines under load, while a commit-once policy that is stable in nominal operation cannot recover work stranded by a failed robot. We present AHE-MRTA, which treats the choice of allocation behaviour as an online decision rather than a design-time one. Its selector, Ecosystem-Driven Paradigm Selection (EDPS), maps four observable context signals—task density, robot availability, deadline pressure, and recent failure rate—onto a small repertoire of fixed, interpretable allocation behaviours, combining deterministic overrides for acute failure and deadline regimes with a dominance dynamic elsewhere; a geodesic-ETA cost oracle and a bounded terminal load-repair step complete the method. We evaluate on three layers of increasing physical realism: a deterministic allocation-only campaign, a stochastic navigation proxy (500 seeds), and closed-loop Gazebo Harmonic/ROS 2 Nav2 experiments with 3, 5, and 10 TurtleBot3 robots (300 runs). The first two layers do not separate the leading policies: AHE-MRTA and Consensus-DBTA co-lead the proxy, with a small mixed-stress edge for AHE-MRTA ($+0.007$, p=0.042). Physical execution does separate them. AHE-MRTA completes every task under robot failure at 3 and 5 robots and attains the highest effective on-time completion under deadline pressure (0.79 at 5 robots and 0.76 at 10, against at most 0.66 and 0.52), cutting the strongest baseline's deadline-violation rate by 38% and 49% while deciding in 0.5–3.4 ms; it pays for this with longer routes and lower workload balance. With five seeds per physical cell these are descriptive operating patterns rather than corrected rankings, and the design does not isolate the selector from other execution-stack differences.

Index Terms

Multi-robot systems, task allocation, online algorithms, biomimetic optimization, ROS 2, Gazebo.

I. Introduction

Multi-robot task allocation (MRTA) decides which robot performs each task and in what order. The decision must respect robot capabilities, task requirements, and operational constraints. During operation, it also depends on execution state. A robot may be delayed, unavailable, or committed to a task whose value or feasibility has changed. Recent reviews cover centralized optimization, distributed and market-based methods, metaheuristics, and learning-based policies [1], [2]. For online MRTA, a low-cost assignment at one instant is not enough. The allocator must preserve useful decisions as tasks arrive, robots become unavailable, deadlines approach, and execution exposes new constraints.

No single strategy is best under all of these conditions, because a choice that helps in one regime tends to hurt in another. Three tensions recur. The first is solution quality against reaction cost: centralized formulations jointly optimize assignment and routing when global state is available, folding timing, workload, distance, energy, and deadline terms into one program [3], but re-solving this coupled problem at every event is costly and can overturn a plan that execution has already begun to act on. The second is global optimality against coordination cost: distributed and market-based schemes replace the global solve with local decisions and explicit messaging—group-based auctions serve delivery requests that arrive over time with time windows [4], and consensus-based allocation gives up some optimality for local coordination, speed, and lower energy use [5]—but the locality that bounds their communication also limits how far they exploit global structure. The third is reactivity against stability: an aggressively re-optimizing policy recovers quickly when a robot fails or a deadline slips, yet it preempts running tasks and breaks feasible routes, whereas a commit-once policy is cheap and stable but strands the tasks of a failed robot. Other online methods take fixed positions on these axes—game-theoretic clustering couples task grouping with allocation [6], self-adaptive methods react to shifts in problem structure [7], and decentralized energy-aware policies weight power draw [8]—but each commits to its position in advance. Because arrival bursts, robot failures, and deadline pressure each reward a different position, a strategy tuned for one regime is exposed in another. This motivates treating the allocation behaviour itself as an online choice rather than a fixed design commitment.

This letter considers online single-task-robot allocation with task arrivals, robot failures, mixed stress, and deadline pressure. The allocator acts at a task arrival, failure, deadline alarm, or replanning event. It can use only the state available at that event. Reassigning a task can reduce estimated travel cost but can also preempt execution, disrupt a feasible route, or move delay to another robot. Keeping an incumbent task can also be harmful after a failure or a sharp reduction in time slack. We therefore use completion and deadline violations as primary outcomes. We report delay, workload balance, recovery, redispatches, preemptions, and decision latency as trade-offs. A strong score in a navigation-independent allocator—one measured where

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every dispatched goal is reached, so navigation cannot fail—does not by itself demonstrate closed-loop navigation or recovery performance. Because methods observe and change the system in different ways, we state the observation model, reallocation event, communication assumptions, and outcome metrics for each comparison.

Our method, *AHE-MRTA*, uses Ecosystem-Driven Paradigm Selection (EDPS). EDPS is an interpretable event-triggered selector, not a new optimizer at every event. Four observable signals—task density, robot availability, deadline pressure, and recent failure rate—update five hormone states. The selected state activates one fixed behaviour: spatial greedy, priority-first, deadline-oriented temporal, commit-once stability-oriented, or orphan-first recovery allocation. Acute failure and deadline regimes have explicit overrides. A fixed dominance dynamic chooses the behaviour in all other cases. The contribution is therefore a transparent rule for switching between known policy families. It does not claim that one family is always best or that the hormone metaphor alone creates a new solver.

The paper makes four bounded contributions:

1) It specifies Ecosystem-Driven Paradigm Selection (EDPS), an event-triggered, interpretable policy coupling fixed context signals to fixed MRTA behaviours.
2) It records decision and execution quantities needed to distinguish allocation quality from navigation and recovery effects.
3) It evaluates the policy across multiple robot-team scales in both a navigation-independent allocation-only simulator and a ROS 2/Gazebo stack.
4) It separates allocation-only, stochastic navigation-proxy, and closed-loop Gazebo evidence, and states the boundary that keeps map-aware ETA and recovery effects distinct from a causal claim about EDPS.

The evaluation has three layers that differ in how navigation is modelled. The *navigation-independent* allocation-only simulation lets every dispatched goal succeed deterministically, so its scores isolate allocation and queue-ordering quality from local-planner noise. The stochastic navigation proxy keeps navigation cost fixed but allows position-dependent goal failure, adding resilience to the measurement. The closed-loop Gazebo/ROS 2 stack runs the full physics and Nav2 pipeline. The separation prevents a navigation or recovery effect from being reported as allocation evidence. The closed-loop benchmark compares full methods; it does not test the causal effect of EDPS alone in the execution stack. That claim requires a matched Gazebo selector ablation. Stating this boundary makes the comparison between the adaptive selector and fixed MRTA baselines reproducible.

II. Problem Formulation

Event-driven state. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be the robot set. A requested task τ is described by

$$\xi_\tau = (p_\tau, t^{\mathrm{a}}_\tau, d_\tau, \pi_\tau, c_\tau),$$

where $p_\tau \in \mathbb{R}^2$ is the goal position, t^{a}_τ is its activation time, $d_\tau \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ is its deadline, $\pi_\tau \in \{1, 2, 3\}$ is its priority, and c_τ is its capability requirement. At event time t_k, $\mathcal{T}_k$ contains every activated task that is not completed or irreversibly cancelled. Robot r has position p^k_r, health state $\phi^k_r \in \{\text{alive}, \text{failed}\}$, navigation status, and an ordered queue q^k_r. The queue contains tasks assigned but not completed, including an in-flight task when one exists.

An allocation event is a task arrival, a robot failure, a deadline alarm, or the replanning timer. Let $\mathcal{R}^{\mathrm{av}}_k$ denote robots that are alive, not navigation-stuck, and have remaining queue capacity. Let $\mathcal{I}_k \subseteq \mathcal{R} \times \mathcal{T}_k$ be the set of healthy in-flight robot–task pairs. These pairs are locked: a new allocation event may change future queue entries but does not cancel an executing Nav2 goal.

Feasible event-level decision. The binary variable $x^k_{r\tau} = 1$ means that task τ belongs to robot r's planned queue after event t_k. The feasible set is

$$\begin{aligned}
\mathcal{X}_k = \Big\{ x \in \{0,1\}^{n \times |\mathcal{T}_k|} :\ & \sum_{r \in \mathcal{R}} x^k_{r\tau} \leq 1, && \forall \tau \in \mathcal{T}_k, && (1)\\
& \sum_{\tau \in \mathcal{T}_k} x^k_{r\tau} \leq Q, && \forall r \in \mathcal{R}, && (2)\\
& x^k_{r\tau} = 1 \Rightarrow \big(c_\tau \subseteq \mathrm{cap}(r)\big), && \forall r, \tau, && (3)\\
& x^k_{r\tau} = x^{k-1}_{r\tau}, && \forall (r,\tau) \in \mathcal{I}_k \Big\}. && (4)
\end{aligned}$$

The first constraint gives a task one owner, the second limits queue length, and the third enforces capability compatibility. The final constraint is the in-flight safety lock used in the execution stack. Failed or navigation-stuck robots cannot receive a new task; their unfinished tasks become eligible for redispatch.

Arrival prediction and admissibility. For a candidate pair (r, τ), let $\hat{a}^k_{r\tau}$ be the predicted arrival time obtained from the current queue endpoint, a queue-length delay surrogate, and travel to p_τ. The predictor uses the cached geodesic ETA of Eq. (20) as its route metric. A pair with no route is infeasible. For a finite deadline, the admission rule is

$$\hat{a}^k_{r\tau} > d_\tau + s^k_{r\tau} \quad \Longrightarrow \quad (r, \tau) \notin \mathcal{F}_k, \tag{5}$$

where $\mathcal{F}_k$ is the eligible-pair set. The pair-specific allowance $s^k_{r\tau}$ is zero for new tasks in the nominal configuration. A bounded positive allowance is available only to old rescue tasks and can be widened in failure mode. Thus, the allocator rejects a pair that is incompatible, unavailable, unreachable, or later than the allowance for that task state.

Event-level surrogate. Minimizing non-negative cost over $\mathcal{X}_k$ alone would admit the trivial all-zero assignment because the ownership constraints are upper bounds. The implementation therefore uses a cardinality-first rule. After masking infeasible pairs and preserving locked incumbents, define

$$\mathcal{X}^\star_k = \arg\max_{x\in\mathcal{X}_k} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}_k} x^k_{r\tau}. \tag{6}$$

Among these maximum-cardinality event plans, the selected paradigm minimizes its shaped cost:

$$x^k \in \arg\min_{x\in\mathcal{X}^\star_k} \sum_{r\in\mathcal{R}} \sum_{\tau\in\mathcal{T}_k} C^k_{r\tau} x^k_{r\tau}, \qquad C^k_{r\tau} = \infty \text{ if } (r,\tau) \notin \mathcal{F}_k. \tag{7}$$

Repeated LSA passes implement this lexicographic rule; tasks for which no feasible capacity remains stay open for a later event. The active paradigm then sets queue order (Section III-E). This is an online event-level surrogate, not an offline optimum for the complete mission.

Cost and assignment stability. The implementation evaluates seven normalized features: arrival time $A^k_{r\tau}$, priority penalty $P_\tau = (4-\pi_\tau)/3$, battery risk B^k_r, relative load L^k_r, failure risk F^k_r, deadline urgency U^k_τ, and navigation/recovery risk R^k_r. For a feasible pair, the common cost is

$$C^k_{r\tau} = w_d A^k_{r\tau} + w_p P_\tau + w_b B^k_r + w_l L^k_r + w_f F^k_r + w_t U^k_\tau + w_r R^k_r + \Psi^k_{r\tau}. \tag{8}$$

Here $A^k_{r\tau} = \min(1, \hat{a}^k_{r\tau}/A_{\max})$ with $A_{\max} = 220$ s. Because $\hat{a}^k_{r\tau}$ is an absolute time, this feature saturates for every candidate pair once the run passes $A_{\max}$; late in a long run the travel-time channel therefore stops discriminating and the decision is carried by the remaining features and by $\Psi^k_{r\tau}$. In the reported runs (makespans of 150–390 s) this affects the tail of the 10-robot cells. The other base features are dimensionless. The correction $\Psi^k_{r\tau}$ collects the deadline and critical-battery penalties, deadline-capability and bid bonuses, incumbent sticky bonus, reassignment penalty, and capacity penalty. This form avoids adding seconds directly to normalized features. EDPS supplies w; the active paradigm can also shape task order and eligibility. A safe incumbent retains base urgency, whereas an unassigned or at-risk task receives nonlinear urgency growth.

Run-level outcomes and timing contract. The run-level objective is to maximize completion and minimize deadline violations without hiding unfinished tasks. Let $\mathcal{T}^{\text{done}} \subseteq \mathcal{T}^{\text{req}}$ be the tasks completed by $T_{\max}$. Formally,

$$\max\ \mathrm{CR} = \frac{|\mathcal{T}^{\text{done}}|}{|\mathcal{T}^{\text{req}}|}, \qquad \min\ \mathrm{DVR} = \frac{|\{\tau : d_\tau < \infty\ \wedge (t^{\text{c}}_\tau > d_\tau\ \vee\ \tau \notin \mathcal{T}^{\text{done}})\}|}{\max\bigl(1, |\{\tau : d_\tau < \infty\}|\bigr)}. \tag{9}$$

The remaining reported quantities—delay, makespan, recovery time, workload balance, redispatches, preemptions, decision latency, messages, and distance—describe the cost of these outcomes. Delay and DVR are evaluated on all requested tasks; an unfinished task is right-censored at $T_{\max}$ for delay and is a deadline miss for DVR (Section IV-H).

For the deployed AHE cycle, the decision-time contract is $\ell_k = t^{\text{publish}}_k - t_k \le 50$ ms. Baseline decision latencies are reported as measured comparison outcomes rather than being silently clipped to this AHE service target. This distinction preserves the real-time requirement while making latency trade-offs visible.

III. AHE-MRTA: Ecosystem-Driven Paradigm Selection

Design principle and event-triggered architecture.

Section I argued that no fixed allocation policy is best across changing regimes. AHE-MRTA answers this not with a new global solver but with an event-triggered *selection hyper-heuristic* that chooses, online, among a fixed library of five well-understood low-level behaviours [9]. At each allocation event it observes the fleet state, the open task pool, and the previous queues, updates its context and internal state, activates one paradigm, and publishes robot-specific queues. Adaptation therefore acts on two coupled layers—*which* behaviour runs, and *how* the shared cost function is weighted—both driven by the same context, so a single interpretable state governs the entire decision. The selector runs no new inter-robot negotiation protocol: the central manager publishes only each robot's own queue, keeping a low-bandwidth interface.

The persistent selector state comprises a five-dimensional dominance vector $D_k \in \mathbb{R}^5_{\ge 0}$, the last accepted paradigm, and the remaining dwell counter for that paradigm. The allocator additionally retains in-flight goals, previous task–robot ownership, and an orphan-task pool. This second group is not selector memory: it is needed to enforce the in-flight lock and reassignment safety rules of Section II. Figure 1 summarizes the information flow from an event to queue publication.

TABLE I
CONTEXT PROTOTYPE VECTORS $V_i \in [0,1]^4$ FOR EACH PARADIGM. COLUMN HEADERS: c_1 TASK-DENSITY (TD), c_2 ROBOT-AVAILABILITY (RA), c_3 DEADLINE-PRESSURE (DP), c_4 FAILURE-RATE (FR). HIGH VALUE = PARADIGM EFFECTIVE WHEN THIS CONTEXT DIMENSION IS HIGH.

Hormone	td	ra	dp	fr
H_SPATIAL	0.7	0.7	0.1	0.1
H_CRIT	0.3	0.5	0.8	0.2
H_TEMP	0.5	0.5	0.9	0.1
H_STAB	0.3	0.3	0.3	0.8
H_RECOV	0.3	0.2	0.2	0.9

A. Context representation

At each allocation event t_k, the observable system state is compressed into the four-dimensional context vector

$$c_k = \left[c_{1,k}, c_{2,k}, c_{3,k}, c_{4,k}\right]^\top \in [0,1]^4. \tag{10}$$

Let $n = |\mathcal{R}|$ be the fleet size and $m_k = |\mathcal{T}(t_k)|$ the number of open tasks. Let $\mathcal{R}_k^{\mathrm{av}}$ denote alive, non-stuck robots with queue capacity, and $\mathcal{R}_k^{\mathrm{f}}$ failed or stuck robots. The components are

$$c_{1,k} = \min\left(1, \frac{m_k}{\max(1,n)}\right), \qquad \text{task density,} \tag{11}$$

$$c_{2,k} = \frac{|\mathcal{R}_k^{\mathrm{av}}|}{\max(1,n)}, \qquad \text{robot availability,} \tag{12}$$

$$c_{3,k} = \frac{|\{\tau \in \mathcal{T}(t_k) : d_\tau - t_k \le \Delta_d\}|}{\max(1,m_k)}, \qquad \text{deadline pressure,} \tag{13}$$

$$c_{4,k} = \frac{|\mathcal{R}_k^{\mathrm{f}}|}{\max(1,n)}, \qquad \text{failure rate} \tag{14}$$

Here, $\Delta_d = 60$ s. Tasks with an infinite deadline do not enter the $c_{3,k}$ count. The first two signals describe demand–supply balance, the third the fraction of tasks with shrinking time slack, and the last the capacity loss requiring recovery. They are computed from the task pool and already published robot-status summaries; EDPS therefore adds no inter-robot communication burden.

Each paradigm $i \in \{1,\ldots,5\}$ has a designer-defined context prototype $V_i \in [0,1]^4$. The prototypes are not learned class centres; they are fixed, interpretable priors on which signal combination supports the associated behaviour. Compatibility with the current context is

$$v_{i,k} = \frac{V_i^\top c_k}{\|V_i\|_2\,\|c_k\|_2} \quad \text{and} \quad \mathbf{v}_k = [v_{1,k},\ldots,v_{5,k}]^\top, \tag{15}$$

with zero compatibility if either vector has zero norm. Table I gives the prototypes.

B. Dominance update and cost weights

The dominance vector is initialized uniformly and kept on the probability simplex at every event: $D_0 = \frac{1}{5}\mathbf{1}$, $D_k \succeq 0$, and $\|D_k\|_1 = 1$. Let N_k^{c} and N_k^{f} be, respectively, the numbers of completed and failed tasks between events k and $k+1$. The compatibility-scaled performance feedback and failure support are

$$\mathbf{p}_k = g_k \mathbf{v}_k, \qquad g_k = \frac{N_k^{\mathrm{c}} - N_k^{\mathrm{f}}}{\max(1, N_k^{\mathrm{c}} + N_k^{\mathrm{f}})}, \qquad \mathbf{b}_k = c_{4,k}\,[-0.3, 0, 0, 0.4, 0.6]^\top. \tag{16}$$

Thus, a failure supports H_RECOV and H_STAB while reducing the dominance of the purely spatial behaviour. The cooperation matrix A and suppression matrix S are sparse. $A_{ij} > 0$ means that paradigm j supports paradigm i; $S_{ij} > 0$ means that j suppresses i. Their nonzero entries are reported in Table II.

The update is

$$\begin{aligned} D_{k+1} = \mathrm{Proj}_{\Delta^4}\Big(&\mathrm{clip}_{[0,1]}\left[\alpha D_k + \eta A D_k - \lambda S D_k \right. \\ &\left. + \beta \mathbf{p}_k + \gamma \mathbf{v}_k + \delta \mathbf{b}_k\right]\Big), \end{aligned} \tag{17}$$

where Proj_{Δ^4} denotes the implementation's non-negative ℓ_1 normalization (with a uniform fallback when the sum is zero), rather than an iterative Euclidean projection. The rule combines interaction-inspired winner-reinforcement and competitor-suppression terms with short-horizon performance and context compatibility [10]. Functionally, it is a fixed, low-dimensional smoothing of paradigm preference: context and recent outcomes nudge the dominance vector, while the cooperation–suppression terms

TABLE II
NONZERO ENTRIES OF COOPERATION MATRIX A AND SUPPRESSION MATRIX S. ALL OTHER ENTRIES ARE ZERO.

Matrix	**From**	**To**	**Value**
A (cooperation)	H_CRIT	H_TEMP	0.20
	H_STAB	H_RECOV	0.20
S (suppression)	H_TEMP	H_SPATIAL	0.30

damp abrupt swings, giving learning-free hysteresis. It is not an online learned model and provides no optimality guarantee; all matrices and coefficients are fixed, and acute failure or deadline regimes do not depend on it—they are handled by the explicit overrides of Section III-D.

a) Weight generation.: EDPS also produces the seven weights of the common cost function. The channels correspond, in order, to travel time, priority, battery risk, queue load, failure risk, deadline urgency, and reassignment/recovery cost. The fixed paradigm–weight map is

$$M = \begin{bmatrix} 0.9 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.9 & 0.5 & 0.5 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.1 & 0.3 \\ 0.1 & 0.1 & 0.1 & 0.9 & 0.9 \\ 0.1 & 0.5 & 0.9 & 0.1 & 0.1 \\ 0.1 & 0.1 & 0.1 & 0.3 & 0.9 \end{bmatrix}. \tag{18}$$

Accordingly,

$$w_k^{\mathrm{eco}} = \mathrm{softmax}\left(\frac{MD_{k+1}}{T_w}\right), \qquad w_k = 0.3w_0 + 0.7w_k^{\mathrm{eco}}, \tag{19}$$

where $T_w = 0.3$. When recovery mode is active, w_k is additionally blended toward w_{rec} with $\theta_k = \min(0.80, 0.50 + 0.60c_{4,k})$. If deadline pressure exceeds 0.5 and recovery mode is inactive, the urgency channel is multiplied by 1.5. These weights are the context-dependent part of the common cost in Section II. The evaluated execution stack has no battery model, so the battery channel carries zero weight in every reported run; it is kept in the cost vector so that the same weight map transfers unchanged to battery-aware deployments.

b) Feasibility and stability.: Every paradigm applies the hard admission gate of Eq. (5) and the single-owner and in-flight constraints. For unassigned or at-risk tasks that have consumed more than 0.7 of their deadline budget, the urgency penalty is increased nonlinearly. A task expected to finish on time with its incumbent owner is exempt. Changing a previous owner requires at least a 10% improvement in estimated arrival; orphan and rescue tasks are exempt from this margin rule. These safeguards prevent event-triggered replanning from becoming gratuitous task migration.

C. Geodesic ETA and bounded load repair

AHE-MRTA uses a cached geodesic cost oracle and a terminal, bounded load-repair step. Neither component changes the EDPS selector or the paradigm library. The cost oracle uses cached geodesic distance d_G on the 0.55 m-inflated occupancy mask of the shared Gazebo map, using an eight-connected free-cell graph. A pair is infeasible if no route exists. For robot r with ordered queue $Q_r = (q_1, \ldots, q_j)$, let $e_r(Q_r) = p_r$ when $j = 0$ and $e_r(Q_r) = p_{q_j}$ otherwise. The implemented ETA surrogate is

$$\hat{a}_{r\tau}^k = t_k + \frac{d_G(e_r(Q_r), p_\tau)}{v_r} + j\,(h + s_\tau), \tag{20}$$

where s_τ is the candidate service time and $h = 22$ s is the fixed per-queued-task navigation allowance. This deliberately inexpensive surrogate does not claim to reconstruct every intermediate route edge. The same endpoint distance oracle is used by the admission gate, paradigm costs, and repair burden calculations.

After the primary plan x^0 is produced, the bounded load-repair step considers local candidates that change the robot only for newly assigned tasks. It therefore never cancels an executing Nav2 goal. Let $M(x)$ be the predicted number of deadline misses, $D_G(x)$ total geodesic distance, $J(x)$ the projected active-robot Jain index, $B_r(x)$ accumulated and committed productive load, and $F(x)$ the maximum predicted remaining completion time. The admissible candidate set is

$$\begin{aligned} \mathcal{N}(x^0) = \big\{x : M(x) \le M(x^0), \quad & D_G(x) \le (1+\epsilon) D_G(x^0), \\ J(x) \ge J(x^0), \quad & \max_r B_r(x) \le \max_r B_r(x^0), \\ F(x) \le F(x^0)\big\}, \qquad & \epsilon = 0.10. \end{aligned} \tag{21}$$

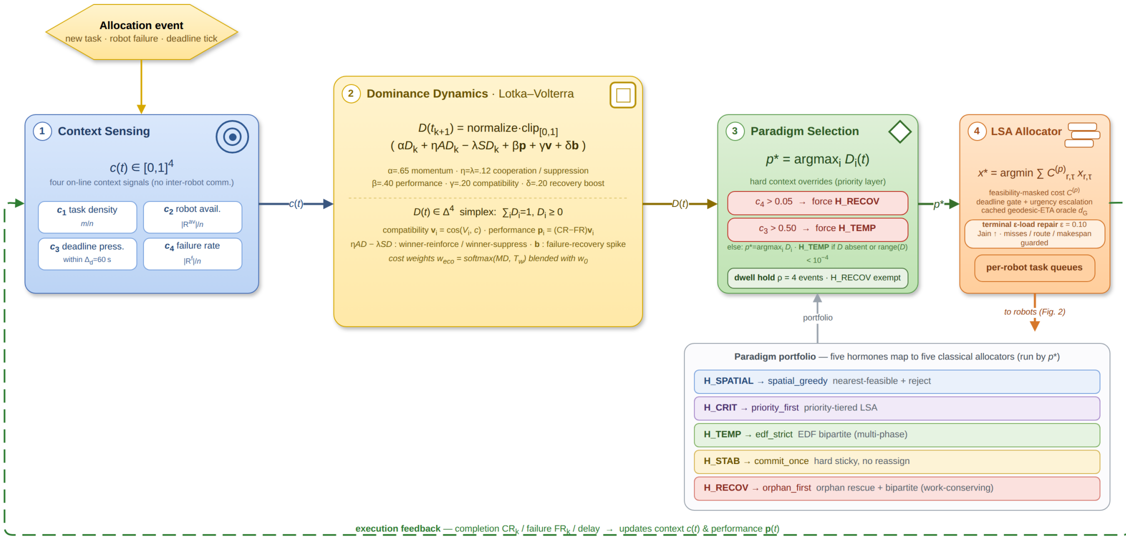


Fig. 1. Ecosystem-Driven Paradigm Selection. Context c_k updates the dominance state; failure/deadline overrides and dwell hysteresis select the active paradigm. The selected paradigm and cost weights produce robot-specific queues through feasibility-masked assignment over the cached geodesic-ETA cost oracle, followed by the terminal bounded load repair. The cooperation and suppression matrices A and S are given in Table II and the fixed coefficients in Section III-G.

Within this set, candidates are ranked by first increasing $J(x)$ and then, in order, decreasing maximum load, load range, and total distance. Repair is enabled only in the terminal phase, when the remaining active-task count is at most three times the healthy-robot count. For an executing first task, $F(x)$ can use Nav2's published remaining-path length instead of a static grid query. These guards prevent the repair from accepting a visually more balanced reassignment that increases deadline misses, route cost, or remaining makespan.

The geodesic ETA and bounded repair change execution cost but leave the EDPS selector unchanged; the closed-loop benchmark therefore does not independently test the causal effect of the selector. This distinction is maintained in the interpretation of the results.

D. Paradigm dispatch and dwell hysteresis

Acute conditions do not wait for the dominance update to respond. The raw choice $\tilde{p}_k$ follows the priority order

$$\tilde{p}_k = \begin{cases} \text{H\_RECOV}, & c_{4,k} > 0.05, \\ \text{H\_TEMP}, & c_{4,k} \le 0.05 \;\wedge\; c_{3,k} > 0.50, \\ \arg\max_i D_{i,k+1}, & \text{otherwise}. \end{cases} \tag{22}$$

Thus, recovery has priority when an event contains both failure and deadline pressure. H_TEMP is the fallback if D_k is absent or its component range is below 10^{-4}, i.e., if the state is not discriminative.

Because the raw choice can switch rapidly, a change other than H_RECOV is held for four allocation events. More specifically, if $\tilde{p}_k \neq p_{k-1}$ while the dwell counter is positive, the dispatcher returns $p_k = p_{k-1}$ and decrements the counter. Once a change is accepted, the counter is reset to $\rho = 4$. H_RECOV bypasses this hold, so a detected failure does not defer the recovery pass to the next event. This mechanism retains override reactivity while limiting ping-pong switching in normal regimes.

E. Paradigm library

The five behaviours share the same feasible set and cost infrastructure; they differ in task-processing order, cost shaping, and treatment of incumbent ownership. Given a feasibility-masked cost matrix $C_k^{(p)}$, the common assignment kernel is

$$x_k^{(p)} \in \arg\min_{x \in \mathcal{X}_k^\star} \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}(t_k)} C_{r\tau,k}^{(p)} \, x_{r\tau}. \tag{23}$$

For an infeasible robot–task pair, $C_{r\tau,k}^{(p)} = \infty$. In the implementation, linear sum assignment (LSA) is repeated across task tiers or processing phases. Equation (23) is therefore the surrogate for one assignment pass, not a claim of global optimality for the entire run.

- **H_SPATIAL / *spatial_greedy*:** Tasks are processed in deadline order and assigned to the eligible robot with the smallest expected arrival time. This follows the nearest-feasible rule used in metric MRTA [11], [12].

TABLE III
FIVE HORMONES, FIVE ALLOCATION PARADIGMS.

Hormone	Paradigm	Inner mechanism
H_SPATIAL	spatial_greedy	nearest-feasible + reject
H_CRIT	priority_first	priority-tiered LSA
H_TEMP	edf_strict	EDF bipartite (multi-phase)
H_STAB	commit_once	hard sticky, no reassign
H_RECOV	orphan_first	orphan rescue + bipartite

Algorithm 1 AHE-MRTA Event Cycle

Require: $\mathcal{R}$, $\mathcal{T}(t_k)$, previous queues, D_k, p_{k-1}, dwell counter h_k; fixed A, S, V, M

$c_k \leftarrow$ ContextVector$(\mathcal{R}, \mathcal{T}(t_k))$
Update the feasibility mask, orphan pool, and in-flight locks
$\mathbf{v}_k \leftarrow (\cos(V_i, c_k))_{i=1}^{5}$; $\mathbf{p}_k, \mathbf{b}_k \leftarrow$ PerformanceAndFailureSupport$()$
$D_{k+1} \leftarrow$ UpdateDominance$(D_k, \mathbf{v}_k, \mathbf{p}_k, \mathbf{b}_k, A, S)$ {Eq. (17)}
$w_k \leftarrow$ GenerateWeights(D_{k+1}, c_k, M) {Eq. (19)}
$\tilde{p}_k \leftarrow$ RawChoice(c_k, D_{k+1}) {Eq. (22)}
$(p_k, h_{k+1}) \leftarrow$ ApplyDwell$(\tilde{p}_k, p_{k-1}, h_k)$
$x_k \leftarrow$ Paradigm$_{p_k}(\mathcal{R}, \mathcal{T}(t_k), w_k)$ {Table III}
if in the terminal repair phase **then**
    $x_k \leftarrow$ EpsilonLoadRepair(x_k, d_G, ϵ)
end if
$x_k \leftarrow$ SingleOwnerAndInFlightLock(x_k)
Publish robot-specific queues and store (D_{k+1}, p_k, h_{k+1})

- **H_CRIT / *priority_first*:** Tasks are partitioned into priority tiers $\pi_\tau = 3, 2, 1$. LSA is applied first to the most critical tier and then to lower tiers, so critical tasks claim capacity first. The design follows the tiered allocation idea in priority-aware auction and consensus strategies [13], [14].
- **H_TEMP / *edf_strict*:** The default temporal behaviour processes tasks in increasing d_τ order and applies the hard deadline gate. The implementation uses a fast pass for near-deadline tasks, a recovery pass for any current orphans, and an LSA pass for the remaining tasks. Queue order may be improved by cheapest-insertion candidates that preserve deadline feasibility. The behaviour applies EDF to deadline-constrained MRTA [15], [16].
- **H_STAB / *commit_once*:** Tasks with a previous owner that remains eligible are kept with that owner. Only new tasks receive a single-pass allocation. This lowers reassignment churn but limits flexibility after a failure, making the stability trade-off explicit.
- **H_RECOV / *orphan_first*:** Tasks left by failed, stuck, or navigation-aborted robots are first redistributed among healthy robots through LSA; remaining tasks then receive temporal allocation. Recovery therefore obtains a dedicated decision pass before new work [12], [17].

F. Algorithm

Algorithm 1 gives the execution order at one EDPS event. A task arrival, robot failure, deadline alarm, or replanning timer triggers the cycle. Only queues are published at the end: D_k, A, S, V, and the full cost matrix remain at the manager node.

G. Configuration

The same selector settings are used at all scenarios and fleet scales: $\alpha = 0.65$, $\beta = 0.40$, $\gamma = 0.20$, $\eta = \lambda = 0.12$, $\delta = 0.20$, $T_w = 0.3$, and $\rho = 4$. The base cost profile is $w_0 = (0.34, 0.10, 0.04, 0.16, 0.10, 0.22, 0.04)$ and the recovery profile is $w_{\mathrm{rec}} = (0.55, 0.04, 0.03, 0.22, 0.09, 0.05, 0.02)$, in the channel order given under *Weight generation*. These quantities are neither relearned nor retuned by event or scenario.

Context, dominance, override, and dwell updates have fixed dimension and are therefore $O(1)$. The dominant cost is the LSA call within a paradigm, whose worst-case complexity for one square assignment pass is $O\big(\max(n, m_k)^3\big)$. The geodesic cache reduces repeated goal queries, and load repair is restricted to a fixed number of local move passes. Computational load thus stems from the selected paradigm and optional route/repair layers rather than from the selector itself.

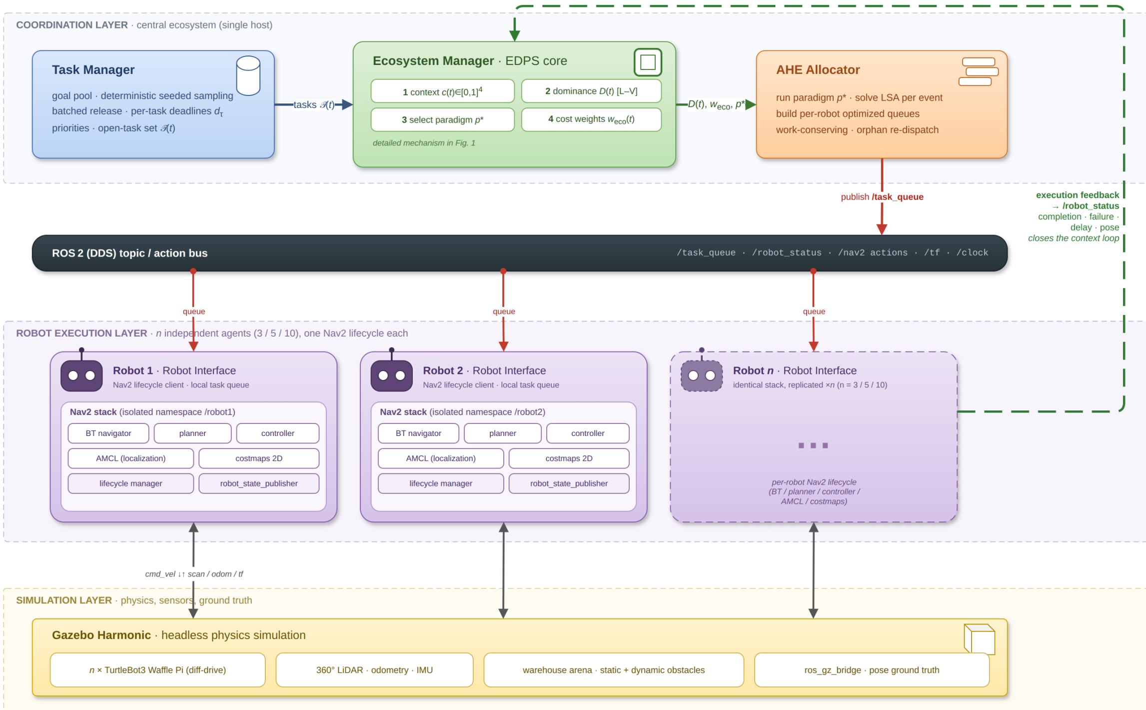


Fig. 2. System architecture. The ecosystem manager updates the hormone state, selects the active paradigm, and publishes one task queue per robot through ROS 2 topics. Each robot runs its own Nav2 lifecycle and returns completion, failure, and delay feedback to the manager.

IV. Experimental Design

A. Hardware Platform

All experiments run on an Intel Core i7-11800H workstation with eight physical cores, 16 threads, a 2.30 GHz base clock, 16 GiB DDR4 RAM, Ubuntu 24.04 LTS, and Linux 6.17. The machine has an NVIDIA RTX 3050 Mobile GPU, but no proprietary driver is installed. Gazebo and RViz therefore use Mesa llvmpipe software rasterization (`LIBGL_ALWAYS_SOFTWARE=1`, `GALLIUM_DRIVER=llvmpipe`). No experiment uses GPU acceleration. The reported comparison is CPU-bound.

B. Software Stack

We use **Gazebo Harmonic** (gz-sim 8.x) [18], **ROS 2 Jazzy** [19], and **Nav2** [20]. Each robot is a **TurtleBot3 Waffle Pi** [21] with a 360-ray two-dimensional lidar (5 Hz, 8 m range), a 200 Hz IMU, and a differential-drive controller ($v_{\max}$=0.26 m/s, $\omega_{\max}$=1.82 rad/s).

Each robot has an independent Nav2 lifecycle in its own ROS 2 namespace. The stack uses global and local costmaps with a 0.40 m inflation radius, a Regulated Pure-Pursuit controller, and a behavior-tree navigator. We provide ground-truth localization through a static `map`→`odom` transform anchored at each robot's spawn pose. This is an experimental control: it isolates allocation effects from localization error (Section VII-F). The `ros_gz_bridge` relays `/scan`, `/odom`, `/cmd_vel`, `/joint_states`, and TF for each robot. A TF relay aggregates the namespaced trees for visualization. All methods use identical Nav2 parameters. Figure 2 shows how the ecosystem manager and the per-robot Nav2 lifecycles are wired together in this stack.

C. Inspection Arena

The arena is a 20×20 m walled indoor mock-up. Two horizontal walls with a 1.5 m central passage create upper and lower lanes. Two rows of square pillars (0.5×0.5 m) line the side walls, and four cylindrical obstacles (r=0.35 m) lie in the lanes. We define 42 fixed inspection waypoints. The waypoints are obstacle-aware and at least 1 m apart. Each run samples its task set deterministically from this grid using its seed. Figure IV-C shows the arena, spawn positions, and sampled task sets.

D. Evaluation Scales

We evaluate three Gazebo scales (Table IV). Each scale contains 60 runs: 4 methods × 3 scenarios × 5 seeds.

At N>3, some per-robot `odom`→`base_link` transforms arrived after costmap activation. We resolve this lifecycle race by delaying activation for robot i by $6i$ s at 3r and 5r and by $20i$ s at 10r. We also broadcast an identity transform at node startup. All scales use the same arena, obstacle map, and Nav2 parameters.

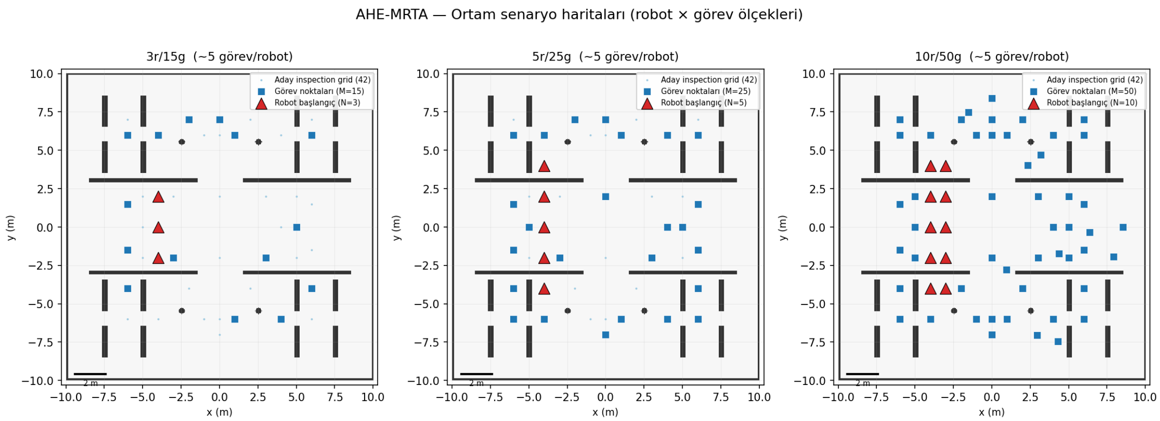


Fig. 3. Arena configurations by scale. **Left:** 3r/15t, with robots in one central column. **Centre:** 5r/25t, with two spawn columns. **Right:** 10r/50t, with all five columns active. Blue stars are task goals for seed 1; colored triangles are robot spawns. All waypoints are obstacle-free.

TABLE IV
GAZEBO EVALUATION SCALES

Scale	Robots	Tasks	Density	Total exps
3r-low	3	9	3.0 t/r	60
3r-mid	3	15	5.0 t/r	60
3r-high	3	24	8.0 t/r	60
5r-mid	5	25	5.0 t/r	60
10r-mid	10	50	5.0 t/r	60
Total Gazebo experiments				**300**

E. Stress Scenarios

The three scenarios stress different allocation requirements:

Every run has the same horizon $T_{\max} = 900$ s; a task still open at $T_{\max}$ is unfinished (Section IV-H). In all three scenarios each task draws an independent priority $\pi_\tau \sim \mathcal{U}\{1,2,3\}$, a service time $\sim \mathcal{U}[2,8]$ s, and a deadline $d_\tau = t_0 + \Delta_\tau$ measured from the run start t_0; the scenarios differ in the deadline budget Δ_τ, the task-arrival schedule, and whether a robot fails.

- **robot_failure:** all tasks are released at t_0 with a nominal budget $\Delta_\tau \sim \mathcal{U}[90,300]$ s. One designated robot is disabled at $t = 45 \pm 5$ s (uniform jitter, drawn per seed), early enough that it is carrying work. Its assigned tasks become orphaned and must be redistributed. This scenario tests reactive recovery and the orphan-rescue behaviour.
- **mixed_stress:** the same failure event is combined with staggered demand and a tighter budget: tasks arrive in three waves at $t = 0$, 30, and 60 s ($\lfloor M/3 \rfloor$, $\lfloor M/3 \rfloor$, and the remainder), and the budget is halved ($\Delta_\tau \sim \mathcal{U}[45,150]$ s). Because deadlines run from t_0, later waves arrive with less slack, so the allocator must absorb arrivals, priority spread, and a failure at the same time.
- **deadline_pressure:** no robot fails and all tasks are released at t_0, but the deadline budget is cut to 40% ($\Delta_\tau \sim \mathcal{U}[36,120]$ s). The lower end of this range is below the Nav2 travel time to the farthest waypoint at $v_{\max}$, so part of the set is unservable on time by construction and the scenario separates urgency-driven dispatch and EDF ordering.

Failure injection targets a fixed robot index rather than a random one, so that the recovery burden is identical across methods for a given seed; task positions, priorities, service times, deadline draws, and the injection time are all seed-controlled and shared by all four methods.

F. Baseline Methods

We compare one operational adaptation from each online-MRTA family represented in the AHE-MRTA library. All adaptations run in the same framework. They share the event loop, ROS 2 topic interface, cost inputs (arrival time, urgency, priority, and load), and Nav2 client. Only the allocation policy differs. The per-robot queue cap Q of Eq. (2) is part of that policy rather than a shared constant: BiG-MRTA and Consensus-DBTA hold Q=5, RoSTAM-EA is unbounded within the event, and AHE-MRTA deliberately runs shallow at Q=2 (raised by one in the terminal repair phase) so that queued work stays reassignable after a failure. We use source-paper parameters where they are specified and otherwise report fixed benchmark choices. The names below thus identify our implementations, not independent reproductions of the complete source systems.

a) BiG-MRTA [22] (bipartite matching).: Our BiG-MRTA adaptation builds a bipartite robot–task graph with fixed multi-criteria edge utility and solves maximum-weight matching at each event. Once assigned, a task is not reallocated. It is the deterministic, lowest-latency, single-paradigm comparison. Its policy avoids replanning but does not recover a task when its assigned robot fails or time slack collapses.

b) RoSTAM-EA [7] (evolutionary).: Our RoSTAM-inspired EA evolves candidate assignment vectors with tournament selection, crossover, and mutation at each allocation event. It dispatches the elite individual. It represents self-adaptive stochastic search. It searches a richer assignment space than one-shot matching, but its decision latency is two to three orders of magnitude higher. Its reoptimization does not encode the cause of a context change.

c) Consensus-DBTA [5] (DBTA2-style consensus bidding).: Each robot shares its two highest-valued task bids; simulated network-wide maximum consensus identifies a winner for each task, and priority rounds prevent one robot from winning conflicting tasks. Its bidding rounds keep assignments largely stable, though it still redispatches after a failure. It represents distributed auction/consensus and is the strongest completion-rate baseline in this study. Re-bidding rounds can delay response to a failure or rapid deadline escalation.

The external baselines are not substitutes for a selector ablation: they differ from AHE in allocation logic as well as paradigm choice. The separate fixed-paradigm proxy ablation in Section VII-D addresses only the selector question in the proxy setting.

d) Comparison scope.: The baseline implementations share the event loop, interfaces, and cost inputs, but they are operational re-implementations rather than independent replications of every source system. The four-method benchmark therefore compares these implementations in one common stack. It is not a universal ranking of the cited algorithms. A controlled Gazebo EDPS ablation against fixed paradigms is required before any closed-loop difference can be attributed to the selector itself (Section VII-F).

G. Evaluation settings

We use three evaluation settings that differ only in how navigation is modelled, so that allocation quality is never conflated with navigation or recovery behaviour. **(i) Navigation-independent** (allocation-only): every dispatched goal is reached deterministically along the shared geodesic map—no local-planner stall, costmap rebuild, or goal failure—and the robot pose advances to each completed goal, while injected robot failures still occur. This setting isolates assignment and queue-ordering quality from navigation noise. **(ii) Stochastic navigation proxy**: navigation cost is a fixed edge, but a goal can fail position-dependently and must be re-attempted, so the metric also rewards resilience to navigation failure. **(iii) Closed-loop Gazebo/ROS 2**: the full physics and Nav2 pipeline above runs end to end. Results from the three settings are tabulated separately and never pooled.

H. Metrics

We report completion rate (CR), deadline violation rate (DVR), makespan, all-task average delay, failure-recovery time, workload balance (Jain), task redispatch rate, execution preemptions, decision latency, and message count. We also report total travel distance where applicable. CR and DVR are the primary outcomes. The other metrics describe efficiency and trade-offs.

a) All-task delay and DVR.: A completed-only average rewards a policy that abandons difficult work. Distant, late, or contended tasks then disappear from the denominator. We avoid this survivorship bias by scoring delay and DVR over *all* requested tasks. Let $\mathcal{T}$ be the requested set, t_τ^{a} the activation time, t_τ^{c} the completion time, and $T_{\max}$ the run horizon. Delay for an unfinished task is right-censored at the horizon:

$$\delta_\tau = \begin{cases} t_\tau^{\mathrm{c}} - t_\tau^{\mathrm{a}}, & \tau \text{ completed}, \\ T_{\max} - t_\tau^{\mathrm{a}}, & \tau \text{ unfinished}, \end{cases} \qquad \bar{\delta} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \delta_\tau, \tag{24}$$

Any task with a deadline is a violation if it completes late *or* never completes:

$$\mathrm{DVR} = \frac{\left|\{\tau : d_\tau < \infty \ \wedge \ (t_\tau^{\mathrm{c}} > d_\tau \ \vee \ \tau \text{ unfinished})\}\right|}{\max\left(1, \left|\{\tau : d_\tau < \infty\}\right|\right)}. \tag{25}$$

We apply this convention to every method and scale. We also record completed-only delay and DVR. The two definitions coincide when CR=1 and differ only when a method leaves tasks unserved.

I. Statistical Analysis

We use pairwise, two-sided Mann–Whitney U tests with Bonferroni correction within each scenario family (three baselines $\times$ seven metrics, or 21 tests per scenario). At the primary 5-robot scale, each cell has n=5 runs (one density and 25 tasks). The 3-robot scale pools three densities (n=15), and the 10-robot scale has n=5. We also report Cliff's δ, an effect-size estimate suited to small cells. The four-method Gazebo cells with n=5 are descriptive. We make no cross-method superiority claim from an uncorrected or non-significant comparison. The 100-seed-per-cell allocation-only campaign and the 500-seed primary navigation-proxy cells are separate controlled tests. Gazebo results describe behavior in the evaluated execution stack, not general deployment performance.

TABLE V
AHE-MRTA NAVIGATION-INDEPENDENT ALLOCATION-ONLY RESULTS (100 SEEDS PER CELL). NAVIGATION SUCCEEDS DETERMINISTICALLY; DISTANCE IS THE GEODESIC PATH LENGTH ON THE SHARED INFLATED OCCUPANCY MAP.

Scale	Scenario	Fit.	CR	DVR	Jain$_{active}$	Jain$_{dist}$	Delay (s)	Distance	Decision (ms)
3r/15t	Robot failure	1.000	1.000	0.000	0.990	0.830	40.1	154.3	0.056
3r/15t	Mixed stress	1.000	1.000	0.000	0.991	0.833	39.9	153.3	0.042
3r/15t	Deadline pressure	1.000	1.000	0.000	0.986	0.965	65.8	155.5	0.042
5r/25t	Robot failure	1.000	1.000	0.000	0.983	0.841	30.6	183.7	0.072
5r/25t	Mixed stress	1.000	1.000	0.000	0.983	0.843	30.7	183.8	0.059
5r/25t	Deadline pressure	1.000	1.000	0.000	0.978	0.953	60.5	235.7	0.052
10r/50t	Robot failure	1.000	1.000	0.000	0.961	0.859	30.9	272.7	0.361
10r/50t	Mixed stress	1.000	1.000	0.000	0.968	0.864	30.9	277.0	0.356
10r/50t	Deadline pressure	1.000	1.000	0.000	0.977	0.946	56.8	436.0	0.259

TABLE VI
ALLOCATION FITNESS (PRIORITY-WEIGHTED ON-TIME COMPLETION), 500 SEEDS, 5 ROBOTS / 25 TASKS. **LEFT:** PERFECT-NAVIGATION PLANE, WHICH ISOLATES ALLOCATION QUALITY—AHE-MRTA REACHES THE OPTIMUM (1.000) IN EVERY SCENARIO, MATCHED ONLY BY CONSENSUS-DBTA AND ABOVE BIG-MRTA (MIXED STRESS) AND ROSTAM-EA (ROBOT FAILURE, MIXED STRESS). **RIGHT:** THE STOCHASTIC NAVIGATION PROXY, WHERE NAV2 GOAL FAILURE (NOT ALLOCATION) SETS THE CEILING; AHE-MRTA AND CONSENSUS-DBTA ARE CO-LEADERS—ROBOT FAILURE AND DEADLINE PRESSURE ARE STATISTICALLY TIED, AND THE ONLY SIGNIFICANT SEPARATION IS MIXED STRESS, WHERE AHE-MRTA LEADS (PAIRED WILCOXON p=0.391/0.042/0.532). **BOLD** MARKS THE BEST PER COLUMN; IN THE STOCHASTIC PLANE ONLY THE STATISTICALLY SUPPORTED LEAD IS BOLDED.

	Perfect nav (alloc. quality)			Stochastic nav (resil.)		
Method	RF	MS	DP	RF	MS	DP
AHE-MRTA*	**1.000**	**1.000**	**1.000**	0.484	**0.487**	0.438
BiG-MRTA	**1.000**	0.974	**1.000**	0.445	0.232	0.441
RoSTAM-EA	0.861	0.864	**1.000**	0.409	0.411	0.426
Consensus-DBTA	**1.000**	**1.000**	**1.000**	0.486	0.481	0.437

V. ALLOCATION-ONLY AND NAVIGATION-PROXY RESULTS

In the navigation-independent setting (Sec. IV-G), the allocator and execution oracle both use geodesic distance on the same inflated occupancy map, so the metrics measure allocation, queue order, and recovery after failure rather than local-planner noise.

Table V shows that AHE-MRTA has fitness=CR=1 and DVR=0 in all nine scale–scenario cells (100 seeds per cell). The active-robot Jain index is 0.986–0.991 at 3r, 0.978–0.983 at 5r, and 0.961–0.977 at 10r. Decision latency is at most 0.361 ms, in the largest 10r/50t cell. During development, six 10r seeds allowed one task to execute on two robots, which inflated CR to 1.02. After adding the in-flight owner lock, we reran those seeds and the complete 10r campaign. CR is then exactly 1.000 in every cell. Because completion and deadline metrics are already at the ceiling, this setting does not separate the leading policies; it establishes that AHE-MRTA attains full, deadline-clean coverage with short geodesic routes.

A. Separate resilience experiment: navigation-proxy

The next experiment simplifies navigation *cost* to a fixed-time edge in the robot–task graph. Navigation *outcomes* remain position-dependent and stochastic: a goal can fail and must be reattempted. This is not a navigation-independent setting. We use it as a navigation proxy for recovery after failed goals. Fitness is the priority-weighted fraction of tasks completed before their deadline. Because a failed goal must be recovered, fitness captures both allocation and resilience to navigation failure. With 500 seeds per scenario, AHE-MRTA reaches optimal allocation fitness in the perfect-navigation setting. In the stochastic proxy it is statistically tied with Consensus-DBTA under robot failure and deadline pressure (paired Wilcoxon $p \geq 0.391$) and holds a small significant edge under mixed stress (+0.007, p=0.042). The residual spread is therefore not an allocation-quality gap (Table VI):

At the primary scale (25 tasks and 5 robots; Table VI, left), AHE-MRTA and Consensus-DBTA both reach the ceiling of 1.000 priority-weighted on-time completion in every scenario. This does not establish an allocation advantage. When navigation becomes stochastic (Table VI, right), all methods fall below the ceiling. AHE-MRTA and Consensus-DBTA remain co-leaders on average, and the largest AHE–Consensus mean gap is 0.007 (0.487 versus 0.481 before rounding), under mixed stress and in AHE-MRTA's favour; paired Wilcoxon p values are 0.391, 0.042, and 0.532 for robot failure, mixed stress, and deadline

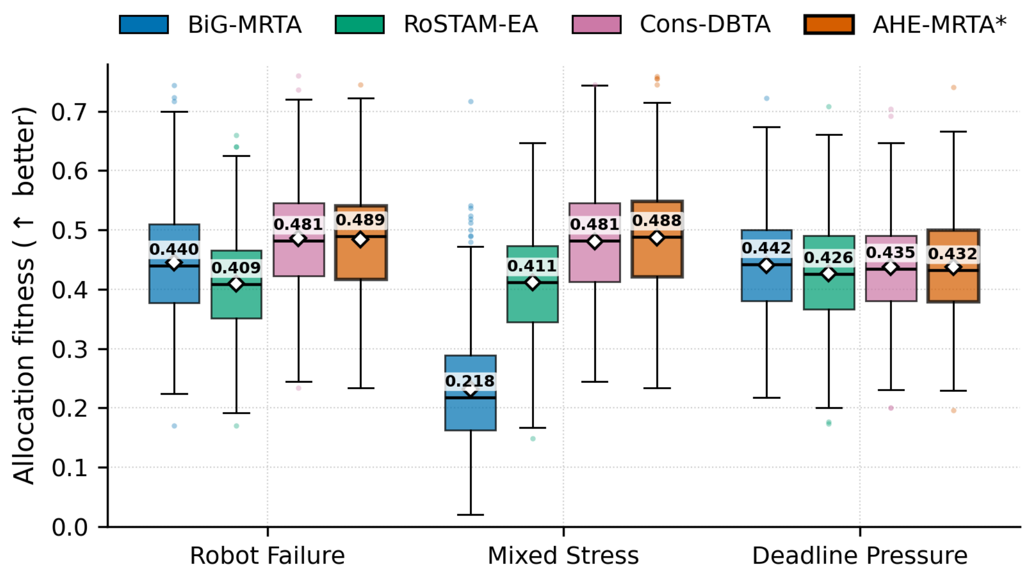


Fig. 4. Navigation-proxy fitness at the primary 5-robot/25-task scale (500 seeds per cell). Boxes show the interquartile range, centre lines the median (value annotated above each box), diamonds the mean, and whiskers 1.5 interquartile ranges; points are outliers. AHE-MRTA and Consensus-DBTA are statistically tied under robot failure and deadline pressure (paired Wilcoxon $p \geq 0.391$); the only significant separation is mixed stress, where AHE-MRTA leads by 0.007 (p=0.042).

TABLE VII

SCALABILITY (STOCHASTIC NAVIGATION PROXY, GEODESIC EXECUTION ORACLE, 100 SEEDS, FIXED DENSITY 5 TASKS/ROBOT, MEAN OVER SCENARIOS). FITNESS ↑, LATENCY (MS) ↓. BEST PER SCALE **BOLD**. PHYSICAL GAZEBO VALIDATION: 3R, 5R, AND 10R CONFIRMED.

	3 robots		**5 robots**		**10 robots**	
Method	Fit.	Lat.	Fit.	Lat.	Fit.	Lat.
AHE-MRTA*	**0.436**	0.13	**0.477**	0.19	**0.455**	0.55
BiG-MRTA	0.363	0.02	0.372	0.04	0.407	0.11
RoSTAM-EA	0.377	2.00	0.420	3.37	0.421	6.77
Consensus-DBTA	0.436	**0.02**	0.476	**0.02**	0.453	**0.04**

pressure. The one cell where a third method is numerically first is deadline pressure, where BiG-MRTA scores 0.441 against AHE-MRTA's 0.438 and Consensus-DBTA's 0.437; that ordering is well inside noise and none of the three separates. Both AHE-MRTA and Consensus-DBTA redispatch failed or orphaned tasks, which is consistent with their similar proxy outcomes. The proxy neither isolates that mechanism nor predicts a closed-loop Nav2 result. We report Gazebo point estimates separately in Section VI. Figure 4 summarizes the proxy results.

B. Navigation-proxy scalability

We examine scalability in two settings. In the navigation proxy, the simplified arrival model permits a high-power sweep over robot count. We run four methods at $N \in \{3, 5, 10\}$ robots, with five tasks per robot ($M = 5N$), 100 seeds per cell, and all three scenarios. This produces 3,600 runs. We also evaluate closed-loop Gazebo execution at 3r, 5r, and 10r. Table VII and Figure 5 average proxy results over scenarios.

AHE-MRTA is tied or numerically highest on proxy fitness at every scale (Table VII). At $N = 3$, it ties Consensus-DBTA (0.436 versus 0.436). The AHE–Consensus differences are only $+0.1$ pp at $N = 5$ and $+0.2$ pp at $N = 10$. We do not interpret these small differences as a performance separation. Proxy decision latency remains about 0.55 ms at 10 robots. This is nearly four orders of magnitude below the 5 s replanning period and about 12 times lower than RoSTAM-EA (6.8 ms at $N = 10$). In the closed-loop Nav2 stack the same allocator decides in 0.5–3.4 ms, once the geodesic oracle is built at start-up rather than inside the first decision (Section VI).

The 5- and 10-robot Gazebo studies contain 60 runs each. Their per-scenario results appear with the 3-robot benchmark in Section VI. They are not a direct confirmation of the proxy because they use a different noise model and fewer seeds.

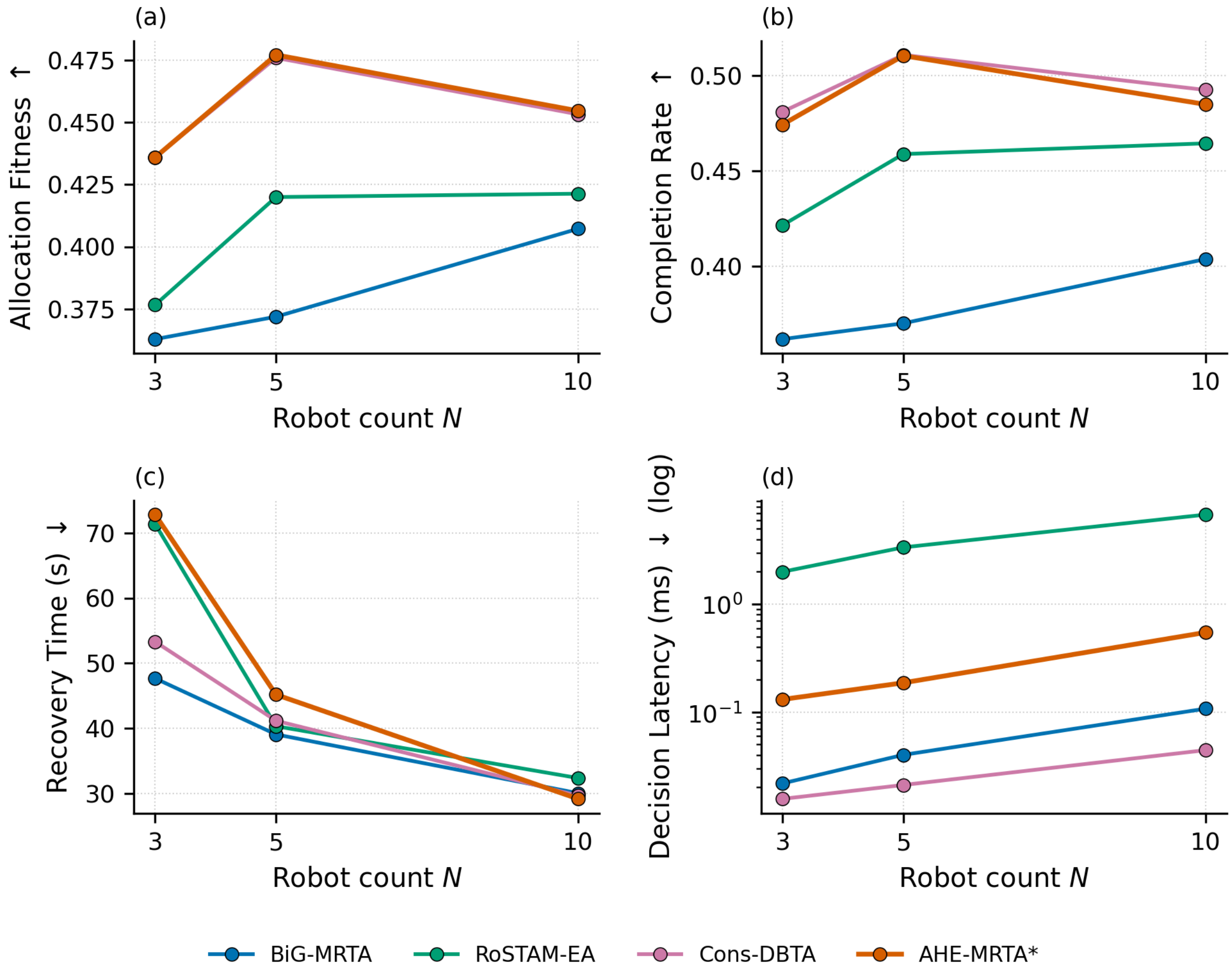


Fig. 5. Scalability by robot count in the stochastic navigation proxy. Values are means over three scenarios with 100 seeds per cell: (a) fitness, (b) completion rate, (c) recovery time, and (d) decision latency. AHE-MRTA is tied with or numerically close to Consensus-DBTA on fitness; the small proxy differences are not treated as separation.

VI. Gazebo Benchmark Results

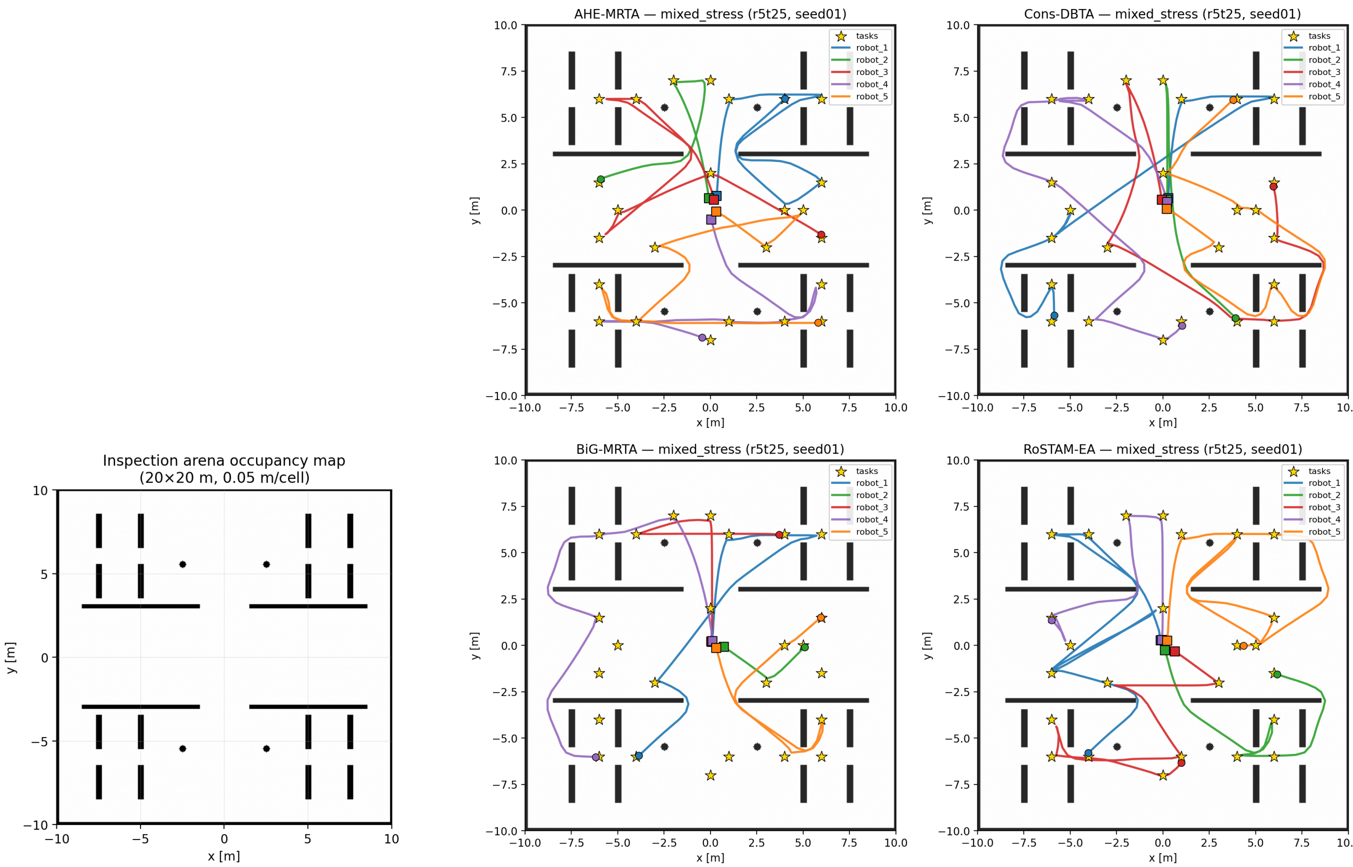
We report the closed-loop Gazebo benchmark. Methods use identical Nav2 parameters, world, and timing budgets; only the allocation policy differs.[1] At 3r, three density levels (9, 15, and 24 tasks) with five seeds per method–scenario combination give 180 runs. The 5r and 10r studies add 60 runs each, for 300 runs in total. We use Bonferroni-corrected Mann–Whitney U tests as described in Section IV-H.

Figure 6 shows the occupancy map and representative trajectories from one *mixed_stress* instance (5 robots, 25 tasks, seed 1). We reconstruct trajectories from logged ground-truth poses and overlay them on the arena and sampled waypoints. In this example, AHE-MRTA and Consensus-DBTA distribute robots across the arena and cover every waypoint. Commit-once BiG-MRTA leaves a larger area uncovered. This is a qualitative illustration, not an aggregate comparison.

Figure 7 shows a mid-run snapshot of the 10-robot configuration under AHE-MRTA. The snapshot comes from an illustrative 10-robot / 50-task run with staggered arrivals and no injected failure; it is outside the three benchmark scenarios and contributes to no reported number. The Gazebo view shows robots travelling to assigned waypoints. The concurrent RViz view shows the occupancy map, each robot's Nav2 global plan, and its ground-truth pose. The snapshot illustrates work-conserving dispatch across both lanes; it is not a quantitative result.

a) Descriptive Gazebo patterns.: Table VIII reports the *robot_failure* and *mixed_stress* cells at the primary scale and Table IX the *deadline_pressure* cell; Figure 8 shows the same scale across all six primary metrics. AHE-MRTA completes every task in the reported runs (CR = 1.000). BiG-MRTA has lower point estimates under *deadline_pressure* (0.728) and *mixed_stress* (0.752). AHE-MRTA* also has the largest reported effective on-time-completion point estimates under deadline pressure: 0.79 at 5 robots and 0.76 at 10 robots. The largest baseline point estimates are 0.66 and 0.52, respectively. These

[1]Tables and figures denote AHE-MRTA as AHE-MRTA* where it is evaluated with its geodesic-ETA cost oracle and terminal load repair (Sec. III-C).



(a) Pre-built Nav2 occupancy map

(b) Executed ground-truth trajectories, four methods

Fig. 6. Experimental map and representative trajectories. **(a)** The static Nav2 occupancy map (`obstacle_map`) is 20×20 m with 0.05 m cells and origin $[-10, -10]$. It contains divider walls, square pillars, and four cylindrical obstacles. **(b)** Paths from one *mixed_stress* run (5 robots, 25 tasks, seed 1) for AHE-MRTA, Consensus-DBTA, BiG-MRTA, and RoSTAM-EA (clockwise from top left). Squares are spawn positions, gold stars are sampled waypoints, and colored curves are paths reconstructed from ground-truth poses. This single-run figure is qualitative; it does not establish an aggregate method ranking.

TABLE VIII

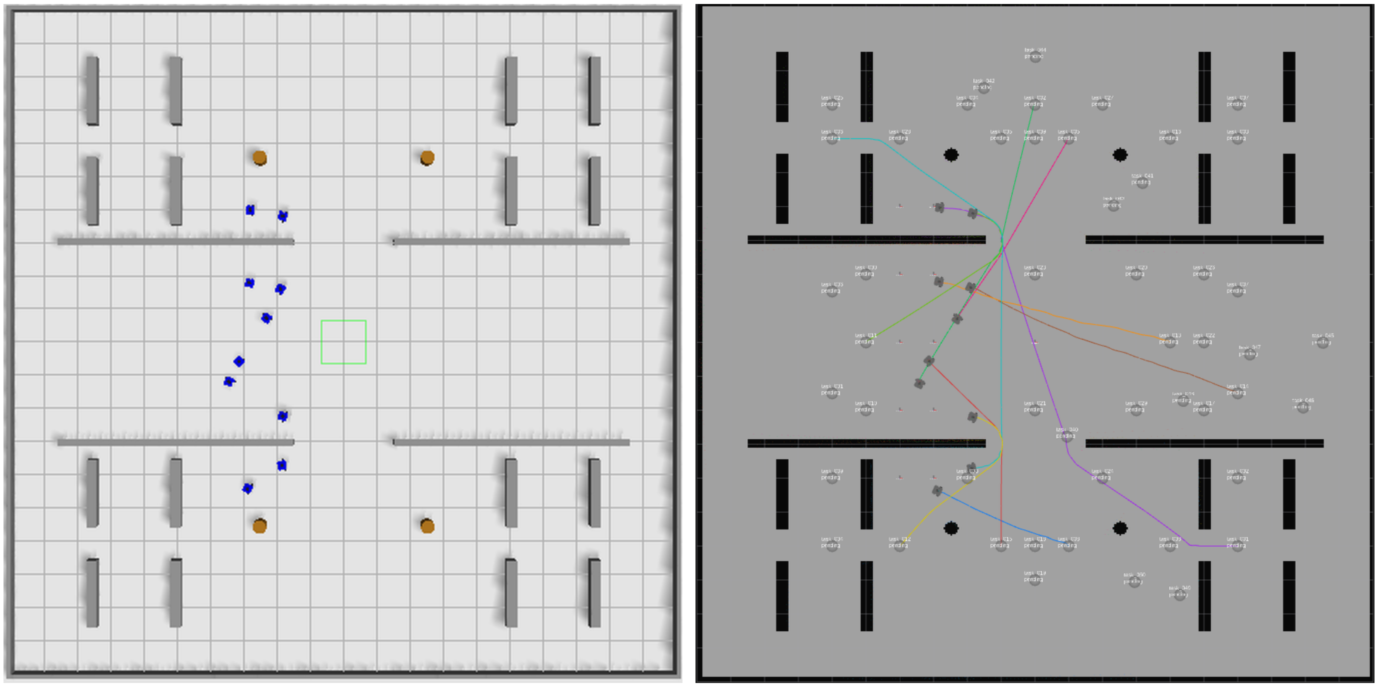
MAIN COMPARISON AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=5 RUNS (SEEDS) PER CELL. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. SIGNIFICANCE VS AHE-MRTA*: *p<0.05, **p<0.01, ***p<0.001 (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN EACH SCENARIO FAMILY OF 21 TESTS).

Method	**CR↑**	**Delay↓(s)**	**RecT↓(s)**	**Preempt↓**	**Churn↓**
Scenario: robot_failure					
AHE-MRTA*	**1.000 ± 0.000**	95.6 ± 12.2	110.9 ± 35.8	0.00 ± 0.00	**0.000 ± 0.000**
BiG-MRTA	0.960 ± 0.049	89.5 ± 40.3	**56.9 ± 33.7**	0.00 ± 0.00	**0.000 ± 0.000**
RoSTAM-EA	**1.000 ± 0.000**	**83.6 ± 11.9**	85.2 ± 34.9	0.00 ± 0.00	0.016 ± 0.022
Cons-DBTA	0.992 ± 0.018	86.4 ± 31.3	178.5 ± 182.9	0.00 ± 0.00	0.048 ± 0.107
Scenario: mixed_stress					
AHE-MRTA*	**1.000 ± 0.000**	54.0 ± 7.7	**27.9 ± 15.4**	0.00 ± 0.00	**0.008 ± 0.018**
BiG-MRTA	0.752 ± 0.059	233.2 ± 42.4	28.4 ± 26.9	0.00 ± 0.00	0.032 ± 0.030
RoSTAM-EA	**1.000 ± 0.000**	**48.3 ± 11.3**	63.9 ± 51.8	0.00 ± 0.00	**0.008 ± 0.018**
Cons-DBTA	**1.000 ± 0.000**	58.7 ± 14.0	59.9 ± 41.9	0.00 ± 0.00	**0.008 ± 0.018**

observations are descriptive. The four-method cells have n=5, and no Bonferroni-corrected cross-method comparison supports a general ranking.

The four-method comparison cannot attribute a Gazebo difference to EDPS because methods differ in several components. The evaluated method is therefore promising under deadline pressure, but a fixed-paradigm/no-override Gazebo ablation is needed before assigning the pattern to paradigm switching.

b) Costs of AHE.: Reactive behavior can increase execution costs. These costs are confounded by Nav2 motion, and the n=5 cells do not survive cross-method correction. Under *robot_failure*, AHE's average delay is 96 s, compared with 84 s and 86 s for the other recovery-capable methods (RoSTAM-EA and Consensus-DBTA) and 89 s for commit-once BiG-MRTA.



(a) Gazebo Harmonic — 3-D physics

(b) RViz — map + live Nav2 plans

Fig. 7. Representative live snapshot of one 10-robot AHE-MRTA run on a illustrative 10-robot / 50-task run with staggered task arrivals, outside the three benchmark scenarios. Both panels are captured from the same paused simulation instant of the same run, so robot positions correspond one-to-one. **(a)** Gazebo Harmonic (top-down view) shows the ten TurtleBot3 Waffle Pi robots travelling in the 20×20 m arena. **(b)** RViz shows the occupancy map, current Nav2 global plan for each robot, and ground-truth-anchored poses. The image illustrates the execution state of one run; it is not comparative evidence.

TABLE IX

DEADLINE SCENARIO RESULTS (DEADLINE_PRESSURE) AT THE PRIMARY 5-ROBOT / 25-TASK SCALE; n=5 RUNS (SEEDS) PER CELL. DVR: DEADLINE VIOLATION RATE. BEST VALUE PER COLUMN IN **BOLD**. SIGNIFICANCE VS AHE-MRTA* (BONFERRONI-CORRECTED MANN-WHITNEY U).

Method	CR↑	Delay↓(s)	DVR↓	Preempt↓
AHE-MRTA*	**1.000 ± 0.000**	71.6 ± 4.3	**0.208 ± 0.033**	0.00 ± 0.00
BiG-MRTA	0.728 ± 0.072	267.0 ± 58.1	0.280 ± 0.057	0.00 ± 0.00
RoSTAM-EA	**1.000 ± 0.000**	70.0 ± 3.1	0.416 ± 0.067	0.00 ± 0.00
Cons-DBTA	**1.000 ± 0.000**	**64.5 ± 8.7**	0.336 ± 0.140	0.00 ± 0.00

Its 111 s recovery time is lower than Consensus-DBTA's 179 s, but higher than RoSTAM-EA's 85 s and commit-once BiG-MRTA's 57 s (Figure 10). Under *mixed_stress* and *deadline_pressure*, BiG-MRTA has the largest delay (233–267 s) because its completed tasks finish very late; this is not a like-for-like comparison when it abandons other tasks. On redispatch churn, AHE is in the best band: at most 0.01 redispatches per task and zero execution preemptions. We count executed reassignments here; the `allocation_instability` counter logged by the runner counts allocator invocations instead, and reports values an order of magnitude larger for every method (e.g., 0.97 for AHE and 4.47 for BiG-MRTA in this cell), so the two must not be compared. Decision latency is not among these costs. The geodesic ETA oracle depends only on the static map and the task pool, both known before the run, so it is built once at start-up instead of inside the first timed decision; AHE-MRTA then decides in 0.8–1.2 ms at 5 robots and 1.9–3.4 ms at 10, overlapping the commit-once baselines (0.10–1.82 ms) and far inside the 50 ms budget.

c) Efficiency.: Table X gives secondary efficiency metrics at the 3-robot scale, where the three densities pool to n=15 per cell. There AHE-MRTA takes 0.47–0.72 ms per decision, against 0.08–0.13 ms for BiG-MRTA, 0.17–0.51 ms for Consensus-DBTA, and 30.8–48.9 ms for RoSTAM-EA. The corresponding figures at the primary 5-robot scale are 0.80–1.24 ms for AHE-MRTA, 0.13–0.30 ms for BiG-MRTA, 0.4–2.4 ms for Consensus-DBTA, and 46–96 ms for RoSTAM-EA; AHE-MRTA therefore stays within the 50 ms budget at every scale while RoSTAM-EA does not.[2] Consensus methods have nearly perfect

[2]The AHE-MRTA latency figures come from a 75-run re-measurement with the geodesic oracle built at start-up, matching the original campaign's design cell for cell (five scale–density cells × three scenarios × five seeds); every other AHE-MRTA quantity and all baseline quantities come from the 300-run campaign. The change moves map-derived set-up out of the timed path and leaves allocation decisions bit-identical, so no other metric is affected: in a paired control at 5r/25t the unmodified arm reproduced the campaign's latency (15.4 versus 15.9 ms) while every non-latency difference stayed inside ordinary Gazebo run-to-run variation.

TABLE X

EFFICIENCY METRICS (GAZEBO, 3-ROBOT SCALE; DENSITIES 9/15/24 POOLED, n=15 PER CELL). WLBAL: ALL-ROBOT JAIN WORKLOAD BALANCE; LAT: MEAN DECISION LATENCY; MSGS: ALLOCATION MESSAGES; DIST: TOTAL TRAVEL DISTANCE. BEST VALUE PER COLUMN (WITHIN SCENARIO) IN **BOLD**. AHE-MRTA* LATENCY IS THE WARM-UP RE-MEASUREMENT AT THE SAME n=15; ALL OTHER ENTRIES COME FROM THE ORIGINAL CAMPAIGN.

Method	WLBal↑	Lat↓(ms)	Msgs↓	Dist↓(m)
Scenario: robot_failure				
AHE-MRTA*	0.853	0.49	32	108.0
BiG-MRTA	**0.949**	**0.13**	131	**72.8**
RoSTAM-EA	0.947	40.24	**2**	84.4
Cons-DBTA	0.891	0.29	12	90.9
Scenario: mixed_stress				
AHE-MRTA*	0.857	0.47	35	113.1
BiG-MRTA	0.928	**0.08**	177	**57.4**
RoSTAM-EA	**0.931**	30.84	**4**	93.7
Cons-DBTA	0.875	0.17	10	104.0
Scenario: deadline_pressure				
AHE-MRTA*	0.986	0.72	25	99.2
BiG-MRTA	0.965	**0.09**	180	**43.9**
RoSTAM-EA	0.978	48.86	**1**	76.7
Cons-DBTA	**0.997**	0.51	6	86.4

TABLE XI

EFFECT SIZES (CLIFF'S δ) OF AHE-MRTA* VS EACH BASELINE ON PRIMARY METRICS. $\delta > 0$ FAVORS AHE-MRTA*. *p<0.05 (MANN–WHITNEY U, BONFERRONI-CORRECTED WITHIN SCENARIO FAMILY).

Metric	vs	BiG	RoSTAM	Cons-DBTA
Scenario: robot_failure				
CR	δ	+0.60	+0.00	+0.20
DVR	δ	+0.40	+0.80	+0.64
RecT	δ	-0.84	-0.44	+0.04
Churn	δ	-0.00	+0.40	+0.20
Scenario: mixed_stress				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	-0.00	+0.12	+0.48
RecT	δ	-0.12	+0.52	+0.44
Churn	δ	+0.52	-0.00	-0.00
Scenario: deadline_pressure				
CR	δ	+1.00	+0.00	+0.00
DVR	δ	+0.68	+1.00	+0.68
RecT	δ	-0.00	-0.00	-0.00
Churn	δ	-0.00	-0.00	-0.00

Positive δ = AHE-MRTA* better; magnitude $|\delta|$>0.47 = large.

workload balance and few messages because they commit assignments. AHE trades some of this balance for reactive changes and the reported deadline outcomes: its all-robot Jain balance is 0.85–0.99 against 0.93–1.00 for the commit-once baselines, and its routes are longer (99–113 m versus 44–94 m at 3 robots) because redispatched tasks are re-approached from a different robot.

d) Effect sizes.: Small cells limit the power of significance tests, so Table XI reports Cliff's δ. Its sign is normalized so $\delta > 0$ favors AHE. Completion effects versus BiG-MRTA are large: $+1.00$ under *mixed_stress* and *deadline_pressure*, and $+0.60$ under *robot_failure*. Deadline-compliance effects under *deadline_pressure* are $+0.68$ versus BiG-MRTA and Consensus-DBTA, and $+1.00$ versus RoSTAM-EA. Under *robot_failure*, they range from $+0.40$ to $+0.80$. The largest unfavorable effect is recovery time versus commit-once BiG-MRTA (-0.84 under *robot_failure*). Recovery is nearly level versus reactive Consensus-DBTA ($+0.04$). On corrected redispatch churn, AHE is comparable or better ($+0.40$ versus RoSTAM-EA and $+0.20$ versus Consensus-DBTA under *robot_failure*). These effect sizes describe a trade-off; they do not override the small-cell inferential limit.

e) Statistical significance.: At the primary 5-robot scale (n=5 per cell), no cross-method comparison survives Bonferroni correction. With n=5, the smallest possible two-sided Mann–Whitney value is about 0.008, above the corrected threshold of about 0.0024. The scale is therefore underpowered; 14 of 63 comparisons reach only raw $p < 0.05$. At 3 robots (n=15 per cell), 13 of 63 comparisons survive correction, 8 of them in AHE-MRTA's favour. At 10 robots (n=5), none survive. We

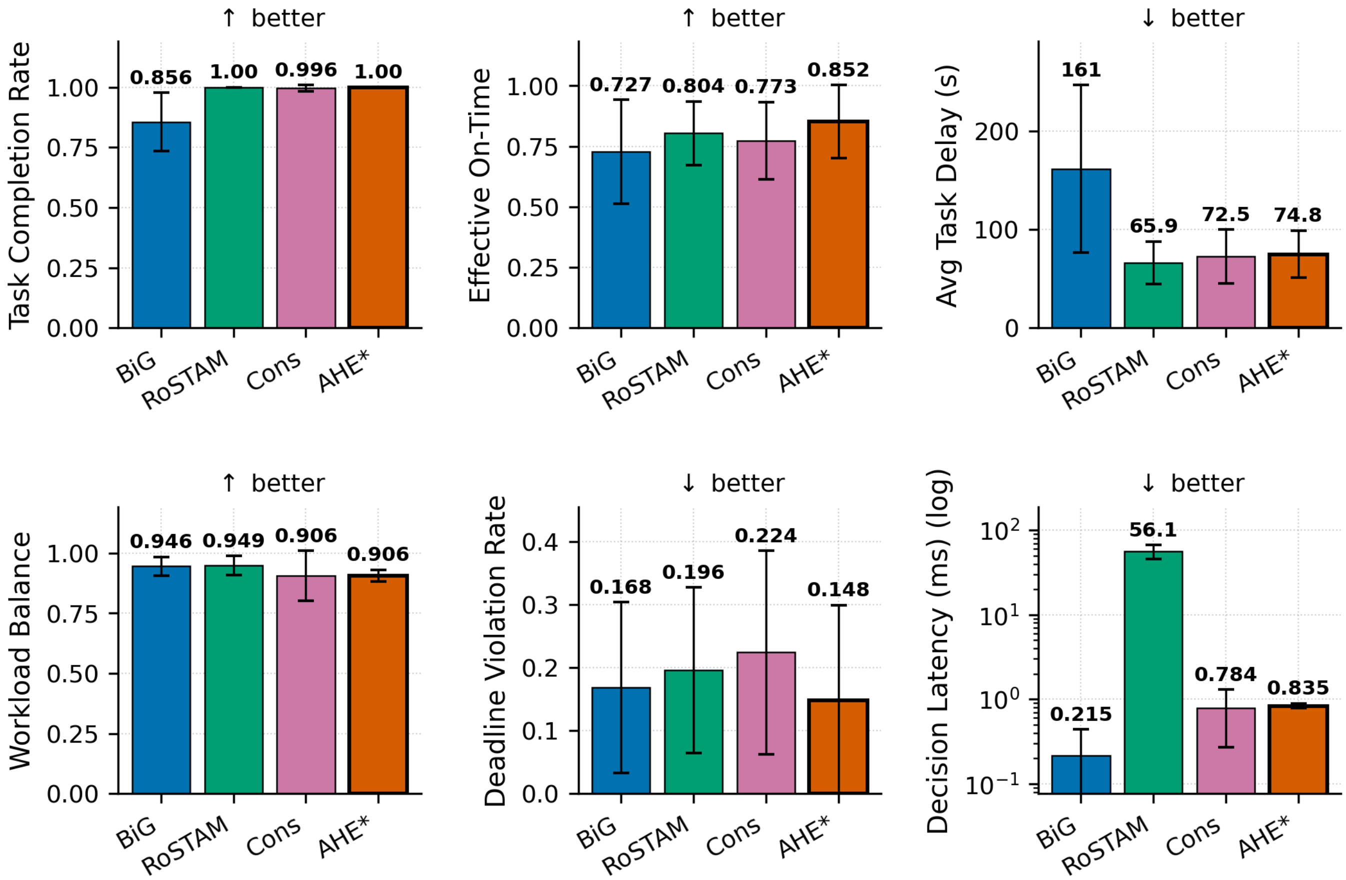


Fig. 8. Multi-metric results at the primary 5-robot/25-task scale (Gazebo Harmonic; *robot_failure* and *mixed_stress* pooled; n=5 seeds per cell; mean ± standard deviation). Panels show completion rate, effective on-time completion $\mathrm{CR}\times(1-\mathrm{DVR})$, average delay, workload balance, deadline violation rate, and log decision latency. These pooled point estimates illustrate trade-offs but do not provide a Bonferroni-corrected method ranking.

treat the Gazebo benchmark as validation under the evaluated execution noise and use the 500-seed allocation-fitness study (Section V) as the primary hypothesis test. Effect sizes support, but do not replace, this distinction.

f) Descriptive Gazebo results at 5 and 10 robots.: Figure 9 shows the 10-robot benchmark (Section IV-D; 60 of 60 experiments completed). AHE-MRTA attains the highest—or tied-highest—completion rate in every scenario (CR = 0.976–0.988). Its clearest lead is under *deadline_pressure*: it reaches CR = 0.980 at DVR = 0.220, giving the highest effective on-time completion ($\mathrm{CR}\times(1-\mathrm{DVR}) \approx 0.76$) far ahead of Consensus-DBTA ($\approx$0.52) and RoSTAM-EA ($\approx$0.48), which reach CR = 0.916/0.964 only by tolerating DVR = 0.428/0.504. Under *mixed_stress* AHE posts the top completion rate (CR = 0.988) but Consensus-DBTA edges it on effective on-time completion (0.650 versus 0.641), both well above RoSTAM-EA (0.529). Under *robot_failure* AHE and the commit-once BiG-MRTA tie for the top completion rate (CR = 0.976), with BiG-MRTA slightly ahead on effective on-time completion (0.936 versus 0.915); the task-abandoning BiG-MRTA records its low DVR only where it completes far fewer tasks in the other two scenarios (CR = 0.696–0.724). At the 5-robot scale AHE reaches full completion in every scenario (CR = 1.000) and leads effective on-time completion throughout: *robot_failure* $\approx$0.98 (DVR = 0.016) vs $\leq$0.92, *deadline_pressure* $\approx$0.79 (DVR = 0.208) vs $\leq$0.66, and *mixed_stress* $\approx$0.72 vs $\leq$0.70; BiG-MRTA abandons tasks under load (CR = 0.728/0.752). Decision latency is 0.8–1.2 ms at 5 robots and 1.9–3.4 ms at 10—overlapping the commit-once baselines and 20–82× below RoSTAM-EA, which grows to 48–107 ms at 10 robots—and far within the real-time budget at both scales.

At the 3-robot scale, Figure 11 traces the mechanism behind these aggregates on *robot_failure*: the dominance state, the seed-averaged completion curve across the injection, and the per-run spread of recovery time, delay, and redispatch. Figure 12 gives the corresponding cumulative-completion curves for *robot_failure* and *mixed_stress*.

VII. Discussion

A. Why does ideal-fitness not transfer linearly to Gazebo?

The deterministic allocation-only simulator (100 seeds per cell) removes Nav2 outcome variance such as local-planner stalls, costmap rebuilds, and passing manoeuvres in narrow corridors. It therefore realizes each accepted route according to the shared geodesic oracle. In Gazebo, two execution effects can reduce an allocation advantage:

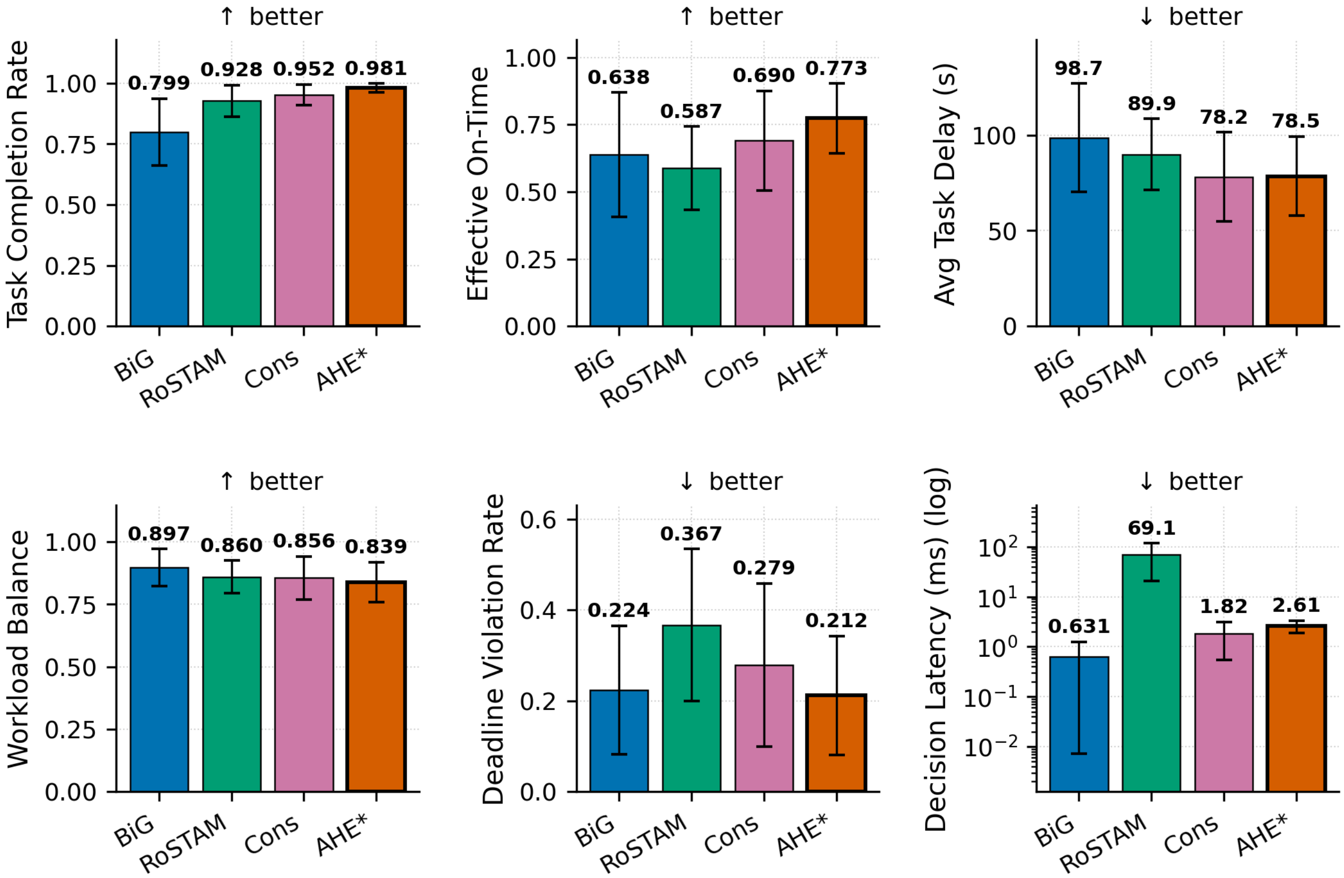


Fig. 9. Multi-metric results at the 10-robot scale (60 runs: four methods, three scenarios, and five seeds; the three scenarios are pooled, so each bar is n=15; mean ± standard deviation; decision latency on a log scale). Pooled over scenarios, AHE-MRTA leads on completion (0.981 versus 0.799–0.952), on effective on-time completion $\mathrm{CR}\times(1-\mathrm{DVR})$ (0.773 versus 0.587–0.690), and is lowest on deadline violation (0.212 versus 0.224–0.367), at 2.6 ms mean decision latency against 0.6–1.8 ms for the commit-once baselines and 69.1 ms for RoSTAM-EA. Pooling hides where the deadline margin actually comes from: it is concentrated in *deadline_pressure* (DVR 0.220 versus 0.296–0.504), which the text reports per scenario. The n=5 per-scenario cross-method cells are descriptive and do not establish a corrected ranking.

1) **Switching has an execution cost.** A paradigm change can reassign a queued task. The new robot must plan and travel to it. A commit-once method avoids this replanning overhead.
2) **Geometry dominates makespan.** After Nav2 dispatches a robot, path length and local motion depend strongly on the map. Extra replanning can increase the longest remaining route.

The Gazebo data is consistent with these effects, but they show up in route cost rather than in the primary outcomes. AHE-MRTA keeps its completion and deadline advantage in the closed loop—it is the completion leader or co-leader in every cell and cuts DVR by 38% (5r) and 49% (10r) against the strongest baseline—while paying for the extra replanning in total travel distance (99–113 m versus 44–94 m at 3 robots) and, at the smaller scales, in makespan under *mixed_stress* (243 s versus 182–219 s at 3 robots). Redispatch churn is *not* where the cost appears: executed reassignments stay at or below 0.01 per task, in the best band of the four methods.

B. Observed operating patterns and attribution boundary

The following per-cell patterns describe the evaluated full configuration. They are not causal estimates of paradigm switching because no selector-only ablation is available:

- **Deadline-compliance pattern.** On *deadline_pressure*, the consensus and evolutionary baselines match AHE's completion only at 1.6–2.3× its deadline-violation rate (5r: AHE DVR = 0.208 vs Cons-DBTA's 0.336 and RoSTAM-EA's 0.416; 10r: AHE 0.220 vs 0.428 and 0.504). AHE-MRTA has the largest effective on-time-completion point estimates at both scales (≈0.79 at 5r, ≈0.76 at 10r). The four-method comparison supports this operating pattern, but not a selector-only explanation.
- **Failure-recovery pattern.** Under *robot_failure* AHE combines full completion at 5 robots (CR = 1.000, DVR = 0.016, effective on-time ≈0.98 vs ≤0.91) with tied-highest completion at 10 robots (CR = 0.976, level with BiG-MRTA on

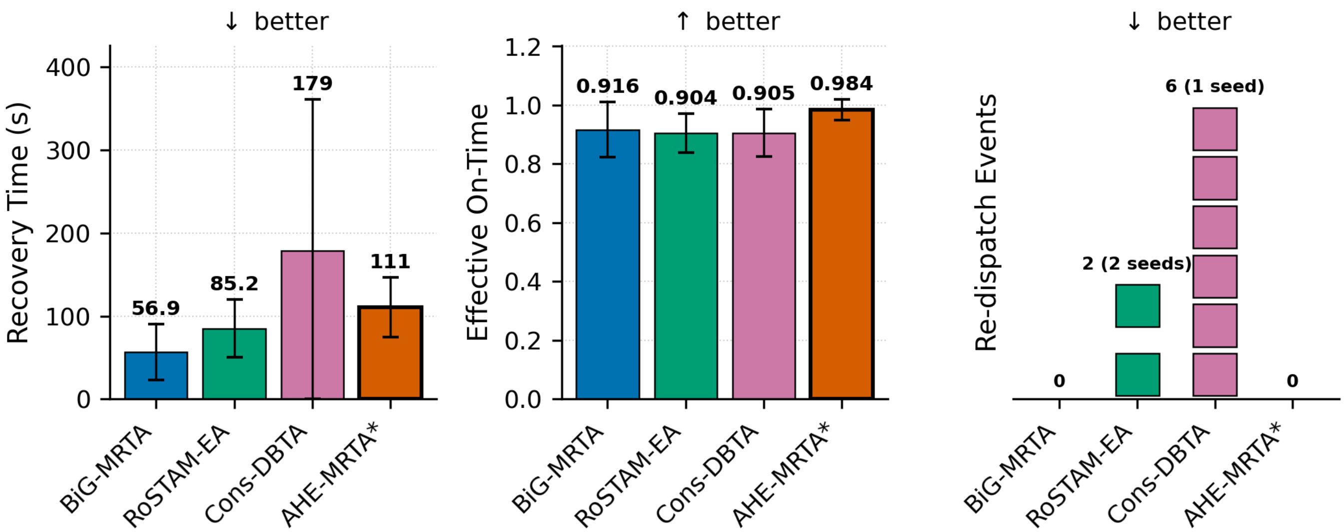


Fig. 10. Failure-recovery behaviour under *robot_failure* at the primary 5-robot / 25-task scale (n=5 seeds): recovery time and effective on-time completion, $\mathrm{CR}\times(1-\mathrm{DVR})$ (mean $\pm$ std bars, whiskers clipped at zero, since neither quantity can be negative), and re-dispatch events as a unit bar in which every square is a single event and squares from different seeds are separated by a gap. This keeps the rare, discrete nature of the metric visible: 17 of the 20 runs have zero re-dispatches, RoSTAM-EA's two events come from two different seeds, and all six Consensus-DBTA events fall in a single seed (execution preemptions are zero for every method and are omitted). Recovery times are statistically indistinguishable across methods (p>0.3). Raw completion rate is omitted because it does not discriminate here—AHE-MRTA and RoSTAM-EA are tied at 1.000, with Consensus-DBTA at 0.992—whereas on effective on-time completion AHE-MRTA leads at 0.984 against 0.904–0.916, i.e. it converts a comparable recovery time into fewer late completions.

effective on-time). The implementation invokes an orphan-rescue behaviour after a failure, but the present four-method comparison does not identify its marginal causal effect.

- **Completion stops separating methods at the small scale, and the residual edge is paid for in makespan.** Under uniform deadline pressure at 3 robots the EDF-style specialists already reach full completion in Gazebo (RoSTAM-EA, Cons-DBTA and AHE-MRTA all at CR = 1.000; only the task-abandoning BiG-MRTA trails at 0.633), so completion carries no information at this scale and the remaining separation is on deadline compliance alone (effective on-time 0.69 for AHE-MRTA versus 0.60–0.66 for the specialists). That margin is bought with a longer makespan (179 s versus 146–159 s), higher delay (80 s versus 70–77 s), and longer routes (99 m versus 77–86 m). The navigation proxy at the same scale likewise puts AHE-MRTA within noise of BiG-MRTA on strict on-time fitness (0.420 versus 0.417, Table VII). At 10 robots, however, AHE keeps DVR at 0.220 against the specialists' 0.428–0.504 and attains the highest effective on-time completion (Sec. VI).

C. Why AHE co-leads rather than dominates the allocation fitness

On the stochastic navigation-proxy fitness (priority-weighted on-time completion), AHE-MRTA and Consensus-DBTA are statistical co-leaders rather than one dominating the other (Table VI). Two mechanisms explain why the strongest methods converge. First, the fitness ceiling is set by *stochastic, position-dependent navigation failure*, not by assignment quality: AHE-MRTA reaches fitness 1.0 in all nine scale–scenario cells of the perfect-navigation allocation-only model (Table V), and the spread in the proxy opens only once goals can fail and must be re-attempted. The binding constraint is therefore resilience to navigation failure—a capability AHE-MRTA and Consensus-DBTA both implement by re-dispatching failed and orphaned tasks—so the two converge to the same frontier, while the commit-once BiG-MRTA, which abandons failed tasks, trails wherever failures accumulate (mixed stress: 0.232 versus 0.481–0.487). It closes the gap only under deadline pressure, where few goals are orphaned and abandoning an already unservable task costs nothing on this metric. Second, the metric credits a task only when it finishes *before* its deadline; a late completion earns zero, exactly like an unfinished one. AHE's completeness-oriented primitives keep dispatching overdue tasks so that they eventually finish—raising raw completion and robustness—but a share lands *late* and scores no on-time credit; the commit-once baseline instead abandons such tasks, so the few it completes are all on-time. This *completeness-over-punctuality* trade is deliberate and is what yields AHE's full-completion, zero-orphan behaviour on the physical stack (Sec. VI). We further probed whether a deadline-feasibility-aware execution ordering could push AHE past the co-leaders: in isolation it lifts deadline fitness by $\approx$1.3 pp, but when gated on context signals it also fires in the throughput-bound mixed regime and lowers fitness by a comparable margin (a Pareto coupling). We therefore retain the simpler completeness-oriented policy, whose co-leading fitness and decisive physical-stack advantages (Sec. VI) are the stronger operating point.

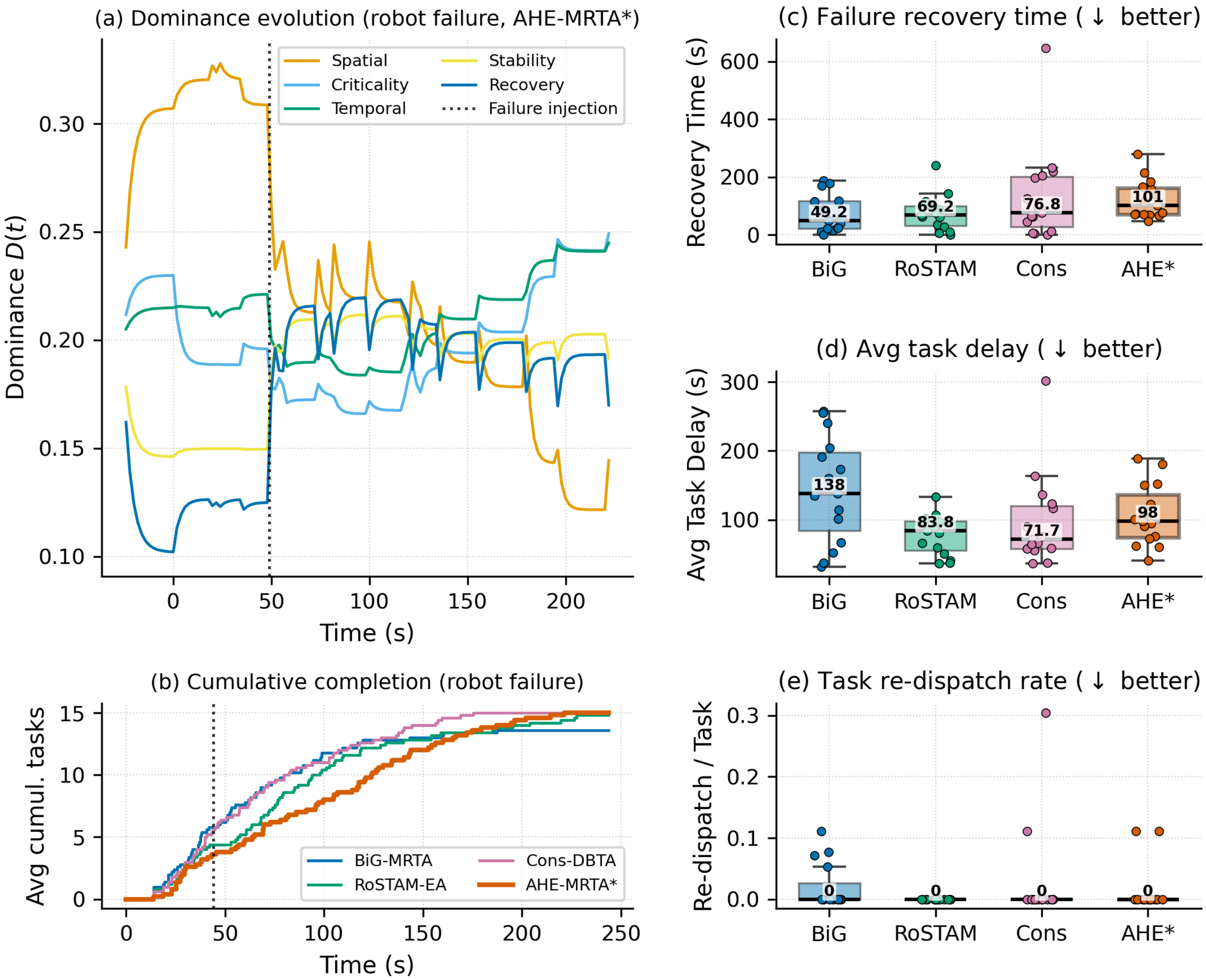


Fig. 11. Interpretable adaptation under *robot_failure* (3r/15t). Time is measured from each run's first task activation, so the injection markers in (a) and (b) sit at the scenario's 45 ± 5 s offset. **(a)** Hormone dominance of a representative AHE-MRTA run; the spatial hormone leads the pre-failure phase ($D{\approx}0.31$), and at the injection (dotted line, $t{\approx}49$ s) it collapses while the recovery and stability hormones rise ($0.12{\to}0.20$ and $0.15{\to}0.20$). No single hormone then regains the pre-failure margin: the criticality and temporal channels lead the remainder of the run and the dispatched paradigm alternates between temporal, spatial, and recovery passes. **(b)** Cumulative task completion (seed-averaged; dotted line: median injection $t{\approx}44$ s): AHE-MRTA continues completing tasks after the failure and reaches the full task set, though it is the last of the three reactive methods to do so. **(c–e)** Recovery time, average task delay, and task re-dispatch rate as box plots (box: IQR; line: median; whiskers: $1.5{\times}$IQR; points: individual runs, $n{=}15$ per method): AHE's reactivity shows up as the highest recovery-time median (101 s versus 49–77 s); its delay median (98 s) sits between the consensus baselines (72–84 s) and the task-abandoning BiG-MRTA (138 s), and the re-dispatch median is zero for all four methods.

D. Ablation: what does the switching machinery add?

To isolate the contribution of online paradigm switching, we replace the EDPS selector with each single *fixed* paradigm, and separately disable the context overrides (*no-override*: pure $\arg\max_i D_i$) and the recovery boost (*no-recovery-boost*: $\delta{=}0$), holding everything else fixed (5r/25t, 100 seeds, stochastic navigation proxy with the geodesic execution oracle).

The eight variants span a mean-fitness range of only $[0.469, 0.479]$ (spread 0.009, or one tenth of the per-seed standard deviation of 0.09): *no* configuration statistically separates from full EDPS, and removing the overrides or the recovery boost leaves proxy fitness essentially unchanged. This proxy ablation establishes only that the tested selector variants do not separate on this navigation-proxy metric. It neither establishes a benefit of EDPS in the physical stack nor rules one out. The four-method Gazebo comparison also cannot fill that gap because its policies differ in more than the selector. Thus, physical differences reported here must be attributed to the evaluated full configurations, not to EDPS alone. A causal test requires matched Gazebo runs of the full configuration against fixed-paradigm, no-override, and no-recovery-boost variants on common seeds and at least one additional map.

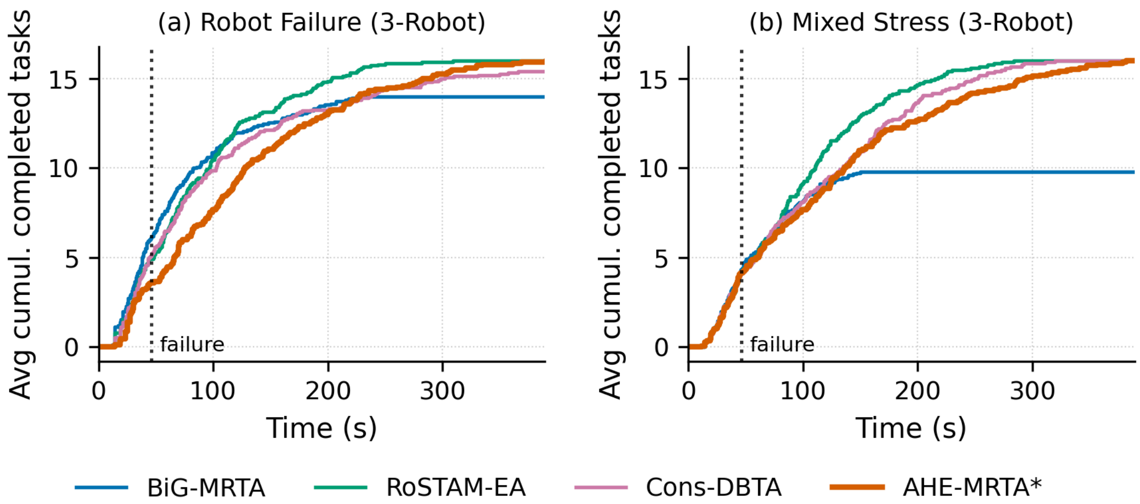


Fig. 12. Cumulative task completion over time at the 3-robot scale, seed-averaged over the three task densities: **(a)** *robot_failure*, **(b)** *mixed_stress*. AHE-MRTA completes the full task set in both scenarios. Its makespan is level with the reactive baselines under *robot_failure* (220 s versus 224 s for Consensus-DBTA) but carries a re-assignment penalty under *mixed_stress* (243 s versus 219 s for Consensus-DBTA and 182 s for RoSTAM-EA). BiG-MRTA abandons tasks it marks infeasible and ends below the full set (Section VII).

TABLE XII
EDPS ABLATION (STOCHASTIC NAVIGATION PROXY, GEODESIC EXECUTION ORACLE, 5R/25T, 100 SEEDS). ALL VARIANTS FALL WITHIN 0.9 PP MEAN FITNESS; NONE STATISTICALLY SEPARATES FROM EDPS.

Variant	**Rob. fail.**	**Deadline**	**Mix. stress**	**Mean**
Full EDPS	0.489	0.445	0.497	0.477
fixed: spatial	0.490	0.448	0.490	0.476
fixed: priority	0.493	0.446	0.486	0.475
fixed: edf	0.481	0.450	0.480	0.470
fixed: commit	0.483	0.437	0.489	0.469
fixed: orphan	0.490	0.451	0.494	0.479
no-override	0.487	0.444	0.497	0.476
no-recovery-boost	0.489	0.445	0.499	0.478

E. Communication footprint

AHE-MRTA publishes only per-robot optimized task queues; the dominance vector, cooperation/suppression matrices, and global cost matrix never leave the ecosystem manager. Figure 13 shows the resulting per-event payload: 84 bytes for AHE-MRTA against 134 and 239 bytes for the commit-once baselines and 1.7×10^4 bytes for RoSTAM-EA, whose population exchange dominates. The interface is therefore a queue broadcast whose size is set by the fleet, not by how often the paradigm switches; we report the payload only and make no claim about its behaviour as a function of switching frequency, which this experiment does not vary.

F. Limitations

(i) **Statistical power at the primary scale:** At the 5-robot primary scale (n=5 per cell) no comparison survives Bonferroni correction—n=5 is structurally underpowered—whereas at 3 robots (n=15) 13 of 63 do; at 10 robots the five seeds per cell are likewise underpowered and no comparison survives correction. We mitigate by reporting the 500-seed stochastic navigation-proxy fitness as the primary high-power cross-method test and the 100-seed-per-cell deterministic campaign as the configuration test, together with Cliff's δ; a larger 10-robot seed budget would add inferential support. (ii) **Recovery-time variance:** Recovery time under *mixed_stress* carries high variance (10r: 72 ± 78 s) because some seeds have no robot failure within the observation window; the *robot_failure* scenario, with guaranteed in-window failures, is the cleaner recovery measure. (iii) **Hardware-bound scale ceiling:** Scaling to 15 robots was infeasible on our 16-thread/16 GiB testbed—multiple per-robot Nav2 stacks failed readiness at launch (startup failures)—so 10 robots is the largest *validated* physical scale. This is a compute limit of the simulation host, not an algorithmic one; the navigation-independent setting confirms the allocation policy itself scales further. (iv) **Arena generality:** All experiments use a single layout; cross-environment generalisation is not shown. (v) **Fixed hormone–paradigm mapping:** The A, S, and V matrices are designer-specified; learning them from domain data

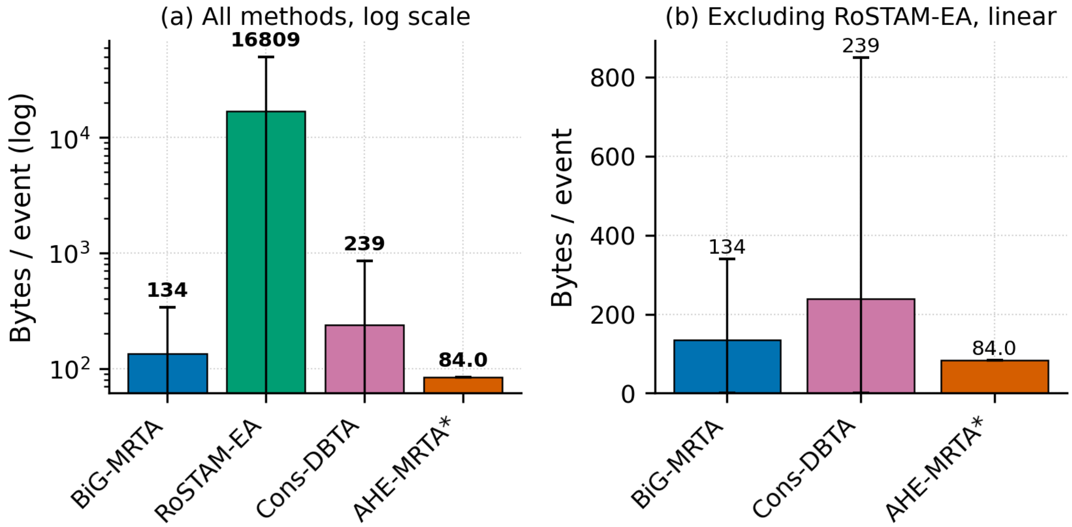


Fig. 13. Communication footprint per allocation event. Only per-robot queues are broadcast; ecosystem internals stay local, keeping bandwidth bounded.

is left to future work. (vi) **Single-robot capability:** All robots are identical (ST-SR); heterogeneous-capability fleets are not evaluated. (vii) **Ground-truth localization:** The physical benchmark supplies pose through a static map→odom transform at the spawn pose, deliberately isolating allocation quality from localization error. Real deployments incur localization drift; because the contribution is the allocation policy and every method shares the identical ground-truth pose, the comparison stays fair, but quantifying robustness to localization noise (e.g., AMCL) is left to future work. (viii) **Bounded role of the hormone dynamics.** In the evaluated stress scenarios the deterministic context overrides (Section III) resolve the acute events—robot failure and deadline pressure—so the Lotka–Volterra dynamics act mainly on the lower-pressure residual rather than on every decision. Replaying the selector over the 9,989 ecosystem states logged across the 75 AHE-MRTA benchmark runs quantifies the split: the failure override decides 50.2% of states, the deadline override 25.0%, and the dominance tier the remaining 24.8%—where it returns spatial-greedy in 99.9% of cases. Replacing that tier with a constant would change the selected paradigm in 3 of 9,989 states (0.03%), and two of the five paradigms, priority-first and commit-once, are never activated in any evaluated scenario. We therefore frame EDPS as a context-driven override cascade *with* an adaptive dynamics tier, not as a dynamics-only selector, and we do not claim the dynamics tier as an operative contributor at these stress levels. The ablation in Table XII (Sec. VII-D) is consistent: removing the overrides or the recovery boost, and replacing the selector with any single fixed paradigm, all leave navigation-proxy fitness within 0.8 pp of EDPS. This proxy observation must not be extrapolated into a closed-loop claim: no matched closed-loop selector ablation is yet available. (ix) **Allocation-fitness model.** The navigation-proxy simulator uses a position-dependent navigation-failure model chosen to *stress* allocation robustness; its failure rate is deliberately higher than the closed-loop Gazebo stack (where Nav2 reaches goals almost always). The proxy therefore measures relative allocation robustness, not an absolute success rate, and is reported alongside—never instead of—the closed-loop Gazebo results. (x) **Missing physical EDPS ablation:** The four-method physical benchmark compares different policies but has $n{=}5$ per cell and does not hold every non-selector component fixed. A decisive follow-up is a common-seed Gazebo experiment of EDPS, each fixed paradigm, no-override, and no-recovery-boost variants, with a pre-specified primary metric, at least 20–30 seeds per condition, and more than one map.

VIII. Conclusion

We presented AHE-MRTA, an event-triggered MRTA framework with fixed, context-driven paradigm selection, a geodesic-ETA cost oracle, and a terminal, bounded load-repair step. In the 100-seed-per-cell allocation-only campaign, AHE-MRTA reaches the completion and deadline ceiling in the tested cells, so this setting does not separate the leading policies. In the stochastic navigation proxy, AHE-MRTA and Consensus-DBTA are statistically tied except for a small mixed-stress edge to AHE-MRTA ($+0.007$, $p{=}0.042$). The two settings provide complementary evidence, not one global performance ranking.

The four-method Gazebo benchmark has five seeds per cell. We therefore report its point estimates as descriptive operating patterns, not as universal superiority. Within the evaluated map and Nav2 stack, AHE-MRTA reaches full completion under robot failure and the largest effective on-time completion under deadline pressure—about 19% (5 robots) and 46% (10 robots) above the strongest baseline, whose deadline-violation rate it also cuts by 38% and 49%—at the cost of longer routes and lower workload balance.

The current experiments do not causally isolate EDPS in closed-loop execution; the external baselines differ in more than the selector. The required next experiment is a matched, multi-map Gazebo ablation of EDPS against fixed paradigms, no-override, and no-recovery-boost variants. It should use a pre-specified primary endpoint and a larger common-seed budget. Separating allocation-only and closed-loop evidence is essential for interpreting the present results.

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